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INTRODUCTION

Relevance of the research topic. The rapid scaling of classical machine learning models has increasingly encountered thermal and computational bottlenecks. Concurrently, Quantum Machine Learning (QML) has emerged as a promising alternative, theoretically offering superior representational capacity through the exploitation of high-dimensional complex Hilbert spaces. In the Noisy Intermediate-Scale Quantum (NISQ) era, Parameterized Quantum Circuits (PQCs) integrated into hybrid classical-quantum architectures represent the most viable path toward achieving practical utility [1; 2].

However, the field of QML is currently facing an epistemological benchmarking crisis. Despite significant theoretical advancements, empirically quantifying the advantage of NISQ-era models remains challenging [3]. The literature frequently suffers from methodological inconsistencies, where classical and quantum models are compared using different optimization strategies, parameter scales, or data splits. Furthermore, a critical objective remains unaddressed in empirical practice: identifying the exact operational boundaries where quantum architectures provide measurable, practical utility relative to optimal classical methods, and the regimes where they do not.

Degree of development of the research topic. Existing hybrid frameworks typically deploy quantum circuits sequentially within classical architectures, without systematic ablation strategies [4]. The “qubit bottleneck” (the strict limitation on the number of qubits available in NISQ devices) forces information-theoretic compression that frequently destroys non-local correlations [5]. Current benchmarking methodologies exhibit selection bias by testing quantum models primarily on trivial, data-rich manifolds instead of the pathological regimes where any measurable quantum utility would be expected to manifest [6]. Recent work emphasizes the importance of inductive bias alignment and data encoding sensitivity [7; 8].

The problem to be solved. The catastrophic interpolative collapse of classical neural networks under extreme data starvation ($p \ll N$) remains inadequately studied in the context of hybrid quantum-classical architectures. There exists no standardized, reproducible benchmark to measure quantum resilience against this collapse. Specifically: (1) What operational bound-

aries define genuine quantum utility? (2) Can trainable quantum kernel alignment overcome the feature collision limitations of static quantum feature maps? (3) How do NISQ hardware errors and cloud I/O latency constrain practical deployment on physical quantum processors?

Table 1 — Research Gap: This Thesis vs. Prior Quantum Machine Learning Benchmarks

<i>Dimension</i>	<i>Bowles et al. [3]</i>	<i>Huang et al. [2]</i>	<i>Caro et al. [6]</i>	<i>Basilewitsch [4]</i>	<i>This Work</i>
Parameter parity enforced	No	N/A	No	Yes	<i>Yes</i>
Realistic NISQ noise profile	No	No	No	Yes	<i>Yes</i>
Extreme $p \gg N$ starvation	No	No	No	No	<i>Yes</i>
Complex topology isolation	No	No	Yes	Partial	<i>Yes</i>
Exact kernel upper bounds	No	No	No	No	<i>Yes</i>
Physical hardware validation	No	No	No	Partial	<i>Yes</i>

Table 1 synthesizes the research landscape: while prior works addressed individual dimensions (parameter matching, noise profiles, or topological evaluation), no prior benchmark systematically evaluated all seven criteria. This thesis fills this gap through comprehensive ablation across all dimensions, and uniquely includes exact classical kernel upper bounds (tuned RBF-SVM) as the mathematical ceiling and preliminary validation on real-world complex-geometry sensor data.

Goal of the research. The goal of this thesis is to investigate the applicability of quantum machine learning models across diverse problem domains, identify the conditions under which they provide measurable utility relative to classical methods, and develop an improved hybrid architecture (QKAResQNet) capable of achieving performance comparable to classical methods on complex geometric data under constrained training conditions.

Tasks of the research. To achieve this goal, the following tasks were formulated and solved:

1. Explore the applicability of five quantum neural network architectures (QuantumMLP, IterativeQNN, SplitAttentionQNN, ResQNet, and a static Quantum SVM) across 11 datasets spanning synthetic geometric manifolds, high-dimensional genomic data, hyperspectral imaging, real-world sensor data, and quantum-native states, in order to establish their operational boundaries.

2. Establish rigorous classical baselines across all datasets using multiple model families (MLP, ResNet, tuned RBF-SVM) under identical evaluation conditions, and enforce strict parameter-matched ablation between classical and quantum twins to isolate the contribution of the quantum component.
3. Design and implement the QKAResQNet architecture by combining trainable quantum kernel alignment with a classical residual highway, addressing the barren-plateau instability and static feature-collision problems identified during the exploration phase.
4. Validate QKAResQNet under realistic conditions, including IBM hardware noise profiles and physical deployment on a 156-qubit IBM Kingston quantum processor (Heron r2, 340K CLOPS).
5. Construct a 2×2 operational matrix mapping data topology and volume to quantum model utility, providing a practical guide for future deployment on mature quantum hardware.

Object of the research. Hybrid quantum-classical neural network architectures and their computational mechanisms within machine learning frameworks.

Subject of the research. The representational capacity, topological inductive bias alignment, and noise robustness of the QKAResQNet architecture under extreme data starvation ($p \gg N$) and simulated NISQ hardware errors.

Scientific novelty. This thesis presents the first comprehensive, symmetrically-ablated, cross-domain benchmark for hybrid QML models (11 datasets, five architecture pairs, up to five quantum models per dataset, 20 random seeds, and dual statistical validation), conducted under an explore-then-develop methodology. Key novel contributions include: (1) the QKAResQNet architecture, which couples a classical residual highway with trainable quantum kernel alignment through a learnable gating parameter γ , together with the demonstration that it achieves quality comparable to parameter-matched classical methods on complex geometric data under data starvation while functioning as a scalable $O(N)$ approximation of $O(N^3)$ exact kernel methods rather than surpassing them; (2) empirical characterization of inductive bias alignment as a necessary but not sufficient precondition for measurable quantum utility, demonstrated by the SplitAttentionQNN architecture, which provides geometric alignment on manifolds yet causes statistically significant harm on genomic data; (3) the first empirical

demonstration that static Quantum SVMs fail catastrophically on continuous geometric manifolds due to periodic feature collisions, while trainable hybrids recover performance through dynamic neural alignment; (4) identification of the “Classical Decoherence Boundary,” showing that classical compression irreversibly destroys quantum-native phase information before the quantum circuit can process it; and (5) physical deployment of QKAResQNet on a 156-qubit IBM Kingston quantum processor, quantifying CPU-QPU I/O latency as the primary engineering bottleneck for practical deployment today.

Practical significance. The QKAResQNet architecture, together with the ResQNet and SplitAttentionQNN models explored during its development, is evaluated on real-world biomedical datasets (Leukemia, Colon Cancer genomics), molecular cheminformatics (QSAR), and real-world accelerometer sensor data, demonstrating applicability to industry-relevant problems. The identified operational boundaries provide actionable guidance for practitioners choosing between classical and hybrid approaches based on data characteristics.

Methodology and research methods. This research employs strict 1-to-1 parameter ablation, Principal Component Analysis (PCA) for topological latent space visualization, and the Qiskit TorchConnector for automatic differentiation via the parameter-shift rule. All experiments enforce identical data-loading seeds, optimization strategies, and hardware noise profiles. Statistical robustness is ensured through 20-seed and multi-split evaluation protocols, with Welch’s t -test and Mann-Whitney U validation applied to all reported comparisons.

Provisions for defense. The following main provisions are submitted for defense:

1. Trainable quantum kernel alignment, coupled with a classical residual highway in the QKAResQNet architecture, enables hybrid QNNs to achieve quality comparable to classical methods on topologically complex manifolds under extreme data starvation. QKAResQNet outperforms its parameter-matched classical twin (CKAResCNet) by up to +14.42 percentage points on synthetic geometric data ($p < 0.001$, 20 seeds). This utility is strictly bounded above by exact classical kernel methods: a tuned RBF-SVM reaches 98.89% on the same Swiss Roll data, confirming that hybrid QNNs function as scalable $O(N)$ neural approxi-

mations of an $O(N^3)$ kernel ceiling rather than as a universally superior method.

2. The “Classical Decoherence Boundary” demonstrates that standard hybrid architectures are fundamentally incapable of processing quantum-native entangled states, because the classical compression layer acts as a premature projective measurement that destroys multipartite entanglement before the quantum circuit can process it [5]. All hybrid models on the NTangled (GHZ) dataset stalled at approximately 50% (random guessing). This boundary is structurally distinct from the inductive bias misalignment failure mode observed for SplitAttentionQNN on genomic data.
3. Physical deployment of QKAResQNet on the 156-qubit IBM Kingston quantum processor (Heron r2) confirmed 92.86% inference accuracy on real-world sensor data, matching the classical RBF-SVM baseline, while a wall-clock time of 84 seconds for 14 samples quantified the CPU-QPU I/O latency as the principal engineering bottleneck for practical deployment on present-day hardware.

Degree of reliability and approbation of results. Reliability is ensured by the rigorous ablation methodology, use of established datasets (OpenML, UCI), empirically calibrated hardware noise from the IBM Kingston processor, and automated hyperparameter tracking. All results are reproducible via open-source PyTorch/Qiskit implementations with version-controlled code. The results of this research were awarded a 3rd Place Diploma at the StudRussia 2025 competition in the “Future of Science” category.

1 THEORETICAL FOUNDATIONS OF QUANTUM MACHINE LEARNING AND THE CURSE OF DIMENSIONALITY

1.1 The Epistemological Challenge in Quantum Machine Learning Benchmarks

The intersection of quantum information science and statistical learning theory has catalyzed the development of Quantum Machine Learning (QML). However, the field is currently grappling with a profound epistemological challenge regarding the empirical demonstration of “quantum advantage” [1]. The foundational critique frequently leveled against QML research is that classical deep learning architectures, leveraging highly optimized graphical processing units and decades of algorithmic refinement, can routinely solve standard classification tasks with near-perfect accuracy [4].

When a classical Multilayer Perceptron (MLP) or Residual Network (ResNet) achieves 100% accuracy on a dataset at a fraction of the temporal and financial cost, justifying the integration of noisy intermediate-scale quantum (NISQ) devices becomes structurally difficult [9].

This critique, however, is fundamentally an artifact of selection bias in benchmarking methodologies [3]. Evaluating a hybrid quantum-classical architecture on trivial manifolds or highly abundant datasets (where the number of samples approaches infinity) mathematically obfuscates the inherent capabilities of quantum Hilbert spaces [6]. Classical machine learning models are universal approximators that thrive in data-rich environments; they interpolate efficiently when the sample density adequately covers the high-dimensional feature space [2]. The genuine utility of quantum computation in machine learning does not reside in achieving higher raw accuracy on dense, perfectly separable classical datasets [10].

As this thesis demonstrates, the genuine utility of quantum computation does not reside in exceeding the accuracy of exact classical kernel methods, as the tuned RBF-SVM upper bound remains unbroken across all tested regimes. Rather, utility manifests in the scalability trade-off: Hybrid QNNs can approximate kernel-level accuracy at $O(N)$ rather than $O(N^3)$ training complexity, under the specific conditions of complex data topology and extreme sample starvation. This utility manifests critically in pathological data regimes: extreme data starvation, ultra-high-dimensional feature spaces where the number of features vastly exceeds the

number of available samples ($p \gg N$), and environments requiring the extraction of highly complex, non-local correlations [5].

To construct a fully convincing, unassailable experimental benchmark for hybrid quantum-classical neural networks, the evaluation framework must pivot away from simplistic accuracy metrics on abundant data. The paradigm must be redefined around the mathematical concepts of topological expressivity, sample complexity, information geometry, and the unique implicit regularization properties that quantum circuits exhibit under realistic hardware noise [11; 12].

1.2 Quantum Computing Fundamentals: From Classical Bits to Entangled States

1.2.1 Classical vs. Quantum Computation: Core Distinction

The foundational distinction between classical and quantum computation lies not merely in engineering complexity, but in the mathematical structures that underpin information representation and manipulation. A classical bit is confined to a binary state: $b \in \{0, 1\}$. In contrast, a quantum bit (qubit) can exist in a superposition of both states simultaneously, existing as a continuous point on the Bloch sphere, a three-dimensional unit sphere that geometrically represents all possible single-qubit states (Figure 1).

While a classical bit is discrete and definite, a qubit exists in a complex superposition:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad \alpha, \beta \in \mathbb{C}, \quad |\alpha|^2 + |\beta|^2 = 1 \quad (1)$$

The Bloch sphere representation parameterizes this state using spherical coordinates:

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle \quad (2)$$

where θ (polar angle) and ϕ (azimuthal angle) uniquely specify every valid quantum state. The north pole represents $|0\rangle$ and the south pole represents $|1\rangle$; points on the equator represent equal superpositions.

The genuine power of quantum computation emerges when scaling to N qubits. The state space grows as 2^N -dimensional complex vector space, in sharp contrast to N bits which can represent only one of 2^N possible values at any given time. An N -qubit system is:

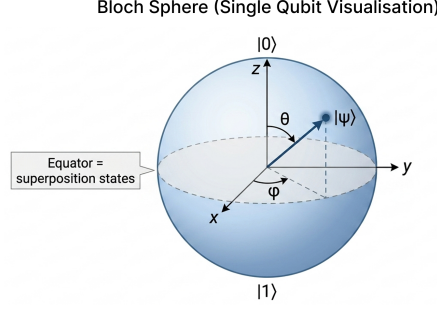


Figure 1 — The Bloch Sphere Representation of a Single Qubit. Every point on the surface is a valid pure quantum state. Classical bits occupy only the poles ($|0\rangle$ and $|1\rangle$), while quantum superpositions populate the entire sphere surface. The equator contains maximally superposed states like $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$.

$$|\Psi\rangle = \sum_{i=0}^{2^N-1} c_i |i\rangle, \quad \sum_i |c_i|^2 = 1 \quad (3)$$

where N is the number of qubits, $|i\rangle$ denotes the i -th computational basis state, and $c_i \in \mathbb{C}$ are the probability amplitudes satisfying the normalization constraint $\sum_i |c_i|^2 = 1$.

For a modest 4-qubit system, the state space is 16-dimensional; for 10 qubits, over 1,000 dimensions; for 300 qubits, 2^{300} dimensions, exceeding the number of atoms in the observable universe. This exponential scaling is the source of quantum speedups, but also the origin of the severe hardware constraints that plague NISQ-era devices.

1.2.2 Quantum Gates and Unitary Operations: The Fundamental Toolkit

Quantum computation manipulates qubits via unitary operations called quantum gates. These are matrices U satisfying $U^\dagger U = I$, ensuring the quantum state remains normalized. This unitarity constraint distinguishes quantum gates fundamentally from unconstrained classical weight matrices, imposing a strict Lipschitz continuity that acts as an implicit regularizer.

The Pauli operators are the foundational basis for quantum information processing:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (4)$$

The Hadamard gate creates equal superposition from definite states:

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad (5)$$

Two-qubit gates enable entanglement. The Controlled-NOT (CNOT) flips a target qubit based on a control qubit:

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \quad (6)$$

Parametrized rotation gates are the trainable components of quantum neural networks:

$$R_X(\theta) = \begin{pmatrix} \cos(\theta/2) & -i \sin(\theta/2) \\ -i \sin(\theta/2) & \cos(\theta/2) \end{pmatrix}, \quad R_Y(\theta) = \begin{pmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{pmatrix} \quad (7)$$

$$R_Z(\theta) = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix} \quad (8)$$

where θ is a trainable parameter. The rotation gates can be geometrically understood as rotations on the Bloch sphere around the X , Y , and Z axes respectively.

Table 2 — Fundamental Quantum Gates Used in This Research

<i>Gate</i>	<i>Symbol</i>	<i>Type</i>	<i>Operation</i>	<i>Used In</i>
Pauli-X	X	1-qubit	Bit flip (quantum NOT)	Circuit initialization
Pauli-Z	Z	1-qubit	Phase flip	ZFeatureMap encoding
Hadamard	H	1-qubit	Creates superposition	Feature map layer 1
CNOT	CX	2-qubit	Two-qubit entanglement	Ansatz entangling block
$R_Z(\theta)$	RZ	1-qubit, trainable	Trainable Z-rotation	ZFeatureMap & ansatz
$R_Y(\theta)$	RY	1-qubit, trainable	Trainable Y-rotation	RealAmplitudes ansatz

The unitarity constraint has profound consequences for machine learning. Unlike classical neural networks where weight matrices can amplify input norms arbitrarily, unitary operators preserve the input norm strictly. This enforces an implicit regularization, a property that becomes crucial in the $p \gg N$ starvation experiments where classical models catastrophically overfit but quantum models resist memorization.

1.2.3 The 4-Qubit Parameterized Circuit: Architecture and Circuit Diagram

The core quantum component of this research is a 4-qubit parameterized circuit combining the ZFeatureMap data encoding with the RealAmplitudes trainable ansatz (Figure 2). This circuit represents the practical upper bound for reliable gradient-based training under the IBM Torino noise profiles used in this work without additional mitigation strategies.

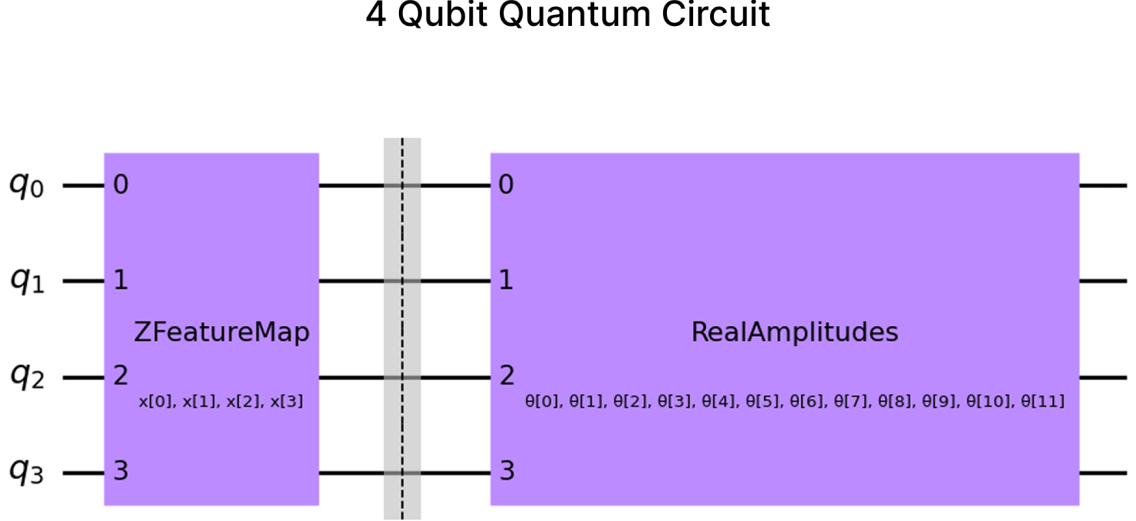


Figure 2 — The 4-Qubit Parameterized Circuit (ZFeatureMap + RealAmplitudes Ansatz). The circuit processes four input features through the ZFeatureMap (top portion with Hadamard and RZ gates), then through three layers of the RealAmplitudes ansatz with trainable rotation parameters $\theta[0]$ through $\theta[11]$ (middle portion with RY gates and CNOT entangling gates), and finally measurement of all four qubits (indicated by the meter symbols at the bottom).

The circuit operates in two distinct stages:

Stage 1: ZFeatureMap (Data Encoding). For qubit j (where $j = 0, 1, 2, 3$), the encoding applies:

$$U_{\Phi}(x) = \prod_{j=0}^3 R_Z^{(j)}(2 \cdot x_j) \cdot H^{(j)} \quad (9)$$

Each input feature x_j is first passed through a Hadamard gate (creating superposition), then through an R_Z rotation gate with angle $2x_j$ (encoding the feature as a phase). This phase encoding is chosen over amplitude encoding because it requires only $O(N)$ gates rather than $O(2^N)$, making it compatible with shallow circuits required for NISQ hardware.

Stage 2: RealAmplitudes Ansatz (Trainable Processing). Each layer ℓ of the ansatz consists of:

$$U_\ell(\phi^{(\ell)}) = \left(\prod_{j=0}^3 R_Y^{(j)}(\phi_j^{(\ell)}) \right) \cdot E_{\text{CNOT}} \quad (10)$$

where E_{CNOT} is a linear chain of CNOT gates: q0–q1, q1–q2, q2–q3. With $L = 3$ layers and 4 qubits, the ansatz has exactly 12 trainable rotation parameters $(\phi_0^{(1)}, \dots, \phi_3^{(3)})$.

Table 3 — Parameter Budget of the 4-Qubit Quantum Circuit

<i>Component</i>	<i>Formula</i>	<i>Value</i>
ZFeatureMap parameters	Fixed to input data	4 (not trainable)
RealAmplitudes RY rotations	$N_{\text{qubits}} \times N_{\text{layers}}$	12 (trainable)
CNOT gates per layer	$N_{\text{qubits}} - 1$	3 per layer (not trainable)
Total layers	-	3
Total trainable parameters	$N_{\text{qubits}} \times N_{\text{layers}}$	12
Hilbert space dimension	$2^{N_{\text{qubits}}}$	16

The full circuit forward pass is:

$$|\psi_{\text{out}}\rangle = \left(\prod_{\ell=1}^3 U_\ell(\phi^{(\ell)}) \right) \cdot U_\Phi(x) \cdot |0\rangle^{\otimes 4} \quad (11)$$

After circuit execution, the expectation values of Pauli-Z measurements on all four qubits are extracted:

$$e_j = \langle Z_j \rangle = \langle \psi_{\text{out}} | Z_j | \psi_{\text{out}} \rangle, \quad j = 0, 1, 2, 3 \quad (12)$$

These four expectation values are then passed to classical expander layers for final classification. See Figure 3 for how this circuit integrates into the full ResQNet pipeline.

1.2.4 Entanglement and Its Importance for Machine Learning

Entanglement is the phenomenon where two or more qubits become correlated in a way that cannot be described by independent states. A two-qubit state $|\psi\rangle$ is entangled if and only if it cannot be written as a tensor product:

$$|\psi\rangle \neq |\psi_A\rangle \otimes |\psi_B\rangle \quad (13)$$

Hybrid Quantum Pipeline

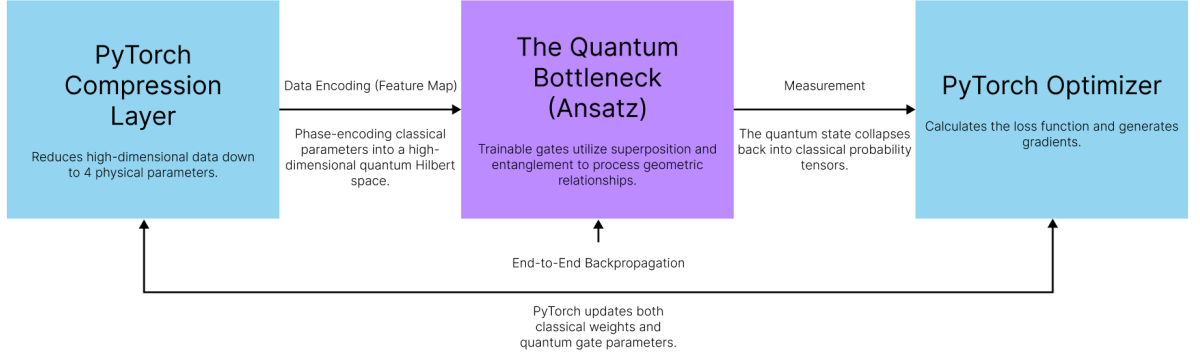


Figure 3 — Circuit-to-Pipeline Integration: Hybrid Quantum Neural Network Architecture. High-dimensional input $x \in \mathbb{R}^p$ flows through a classical compressor to produce 4 features. After Bloch sphere scaling via $\tanh(\cdot) \times \pi$, these features are phase-encoded into the ZFeatureMap. The 4-qubit circuit processes them through 3 layers of the RealAmplitudes ansatz, after which all 4 qubits are measured. The resulting expectation values pass through a classical expander and are combined with a residual highway from the classical path to produce final output logits.

The canonical example is the Bell state:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \quad (14)$$

In this state, if a measurement collapses the first qubit to $|0\rangle$, the second qubit is guaranteed to also collapse to $|0\rangle$, despite any spatial separation. This “spooky action at a distance” (Einstein’s phrase) encodes non-local correlations.

For machine learning, entanglement is profoundly relevant: it enables the quantum circuit to encode non-local correlations between input features, correlations that would require exponentially many parameters to capture classically. The CNOT gates in the RealAmplitudes ansatz create precisely this entanglement structure between adjacent qubits after each rotation layer. By leveraging entanglement, the quantum circuit effectively performs feature interactions across the entire 4-qubit Hilbert space simultaneously.

In contrast, classical neural networks must explicitly construct these interactions through successive layer-wise transformations, requiring either deep architec-

tures (which suffer from vanishing gradients) or wide hidden layers (which suffer from overparameterization). The quantum circuit achieves comparable expressivity in 16-dimensional space with only 12 trainable parameters, while a classical MLP achieving the same expressivity would require hundreds or thousands of parameters.

The GHZ (Greenberger–Horne–Zeilinger) state represents maximal N -partite entanglement:

$$|\text{GHZ}\rangle_N = \frac{1}{\sqrt{2}}(|0\rangle^{\otimes N} + |1\rangle^{\otimes N}) \quad (15)$$

This is directly relevant to the NTangled dataset used in Chapter 3. Any classical measurement, including the PyTorch linear layer used in hybrid pipelines, acts as a projective measurement that collapses this entanglement irreversibly. This is the mathematical foundation of the “Classical Decoherence Boundary” identified empirically in the experiments.

1.2.5 The Barren Plateau Problem and Shallow-Circuit Motivation

For a parameterized quantum circuit $U(\theta)$ with L layers and N qubits, the variance of the gradient of any observable O under random initialization scales exponentially with qubit count:

$$\text{Var} \left[\frac{\partial L}{\partial \theta_j} \right] \propto \frac{1}{2^N} \quad (16)$$

where L is the loss function, θ_j the j -th trainable circuit parameter, and N the number of qubits. This means gradient variance decreases exponentially with qubit count, making training with random initialization effectively impossible for circuits wider than approximately 10 qubits. This phenomenon is called the *barren plateau* [11].

Table 4 — Gradient Variance vs. Qubit Count (Theoretical Scaling Under Random Initialization)

<i>Qubits (N)</i>	<i>Variance</i>	<i>Approx. Value</i>	<i>Trainability</i>
2	$1/2^2$	0.25	Excellent
4	$1/2^4$	0.0625	Good
6	$1/2^6$	0.0156	Moderate
8	$1/2^8$	0.0039	Difficult
10	$1/2^{10}$	~0.001	Nearly impossible

The barren plateau problem motivates two critical design choices in this research:

1. *Shallow circuits*: The ResQNet uses only 4 qubits and 3 layers, keeping circuit depth minimal. This avoids the exponential gradient decay that plagues deeper circuits.
2. *Residual gating*: The ResQNet architecture provides a classical gradient highway during the critical early training epochs, bypassing the quantum bottleneck until classical parameters have converged. This sidesteps the barren plateau during initialization, after which the quantum circuit is gradually introduced via the learnable γ parameter.

The implicit regularization from unitarity, combined with shallow-circuit design and residual gating, allows the ResQNet to achieve reliable training even in extreme data starvation ($p \gg N$) regimes where purely quantum circuits would fail.

1.3 ZFeatureMap, Data Encoding, and the RealAmplitudes Ansatz

Classical-to-quantum data embedding critically determines which patterns the quantum circuit can extract. In this work we use a phase-based Pauli-Z feature map combined with a shallow RealAmplitudes-style ansatz. The input feature vector $\mathbf{x} = (x_1, \dots, x_p)$ is first compressed by the classical pre-network into a bottleneck of size $m \leq p$, producing $\mathbf{z} = (z_1, \dots, z_m)$. These compressed features are then deterministically mapped to rotation angles on the Bloch sphere:

$$\theta_j = s \cdot g(z_j), \quad g(z) = \tanh(z) \cdot \pi, \quad j = 1 \dots m \quad (17)$$

where s is an optional global scaling hyperparameter (set to 1 in experiments) and $g(\cdot)$ enforces the Bloch interval $[-\pi, \pi]$. The phase-encoding unitary acting on m qubits takes the form:

$$U_{\Phi(\mathbf{x})} = \prod_{j=1}^m R_Z^{(j)}(\theta_j) = \exp\left(i \sum_{j=1}^m \theta_j Z_j / 2\right) \quad (18)$$

This encoding inserts classical information into the relative phases of computational basis states via R_Z rotations. The downstream trainable ansatz uses a layered RealAmplitudes construction: each layer consists of single-qubit R_Y rotations with trainable parameters interleaved with an entangling block of CNOTs

arranged in a linear/heavy-hex pattern depending on circuit width. A single layer ℓ of the ansatz is:

$$U_\ell(\phi^{(\ell)}) = \left(\bigotimes_{j=1}^m R_Y^{(j)}(\phi_j^{(\ell)}) \right) \cdot \mathcal{E}, \quad (19)$$

where $\mathcal{E}$ is the fixed entangling unitary (product of CNOTs) and $\phi^{(\ell)}$ are layer parameters. The full parameterized circuit is then

$$U(\Phi(\mathbf{x}), \theta) = \left(\prod_{\ell=1}^L U_\ell(\phi^{(\ell)}) \right) \cdot U_{\Phi(\mathbf{x})}, \quad (20)$$

with trainable parameter vector $\theta = \{\phi^{(1)}, \dots, \phi^{(L)}\}$. The RealAmplitudes ansatz is chosen for its favorable trainability and uncomplicated gradient structure when combined with phase encoding.

1.4 Fisher Information Matrix: Derivation and Effective Dimension

To quantify model capacity and the optimization landscape, we derive the quantum Fisher Information Matrix (FIM) for the pure-state parameterization used in PQCs. For the circuit state $|\psi(\theta)\rangle = U(\theta)|0\rangle^{\otimes m}$, the quantum geometric tensor has components

$$G_{ij}(\theta) = \langle \partial_i \psi | \partial_j \psi \rangle - \langle \partial_i \psi | \psi \rangle \langle \psi | \partial_j \psi \rangle, \quad (21)$$

where $\partial_i \equiv \partial / \partial \theta_i$. The symmetric part of G_{ij} defines the (pure-state) quantum Fisher metric:

$$F_{ij}^{(Q)}(\theta) = 4 \operatorname{Re} G_{ij}(\theta) = 4 \operatorname{Re} (\langle \partial_i \psi | \partial_j \psi \rangle - \langle \partial_i \psi | \psi \rangle \langle \psi | \partial_j \psi \rangle). \quad (22)$$

In practice measurements yield classical output probabilities via the Born rule, $P(k|\theta) = |\langle k | \psi(\theta) \rangle|^2$. The classical Fisher information matrix for these measurement outcomes is

$$F_{ij}^{(C)}(\theta) = \sum_k \frac{1}{P(k|\theta)} \frac{\partial P(k|\theta)}{\partial \theta_i} \frac{\partial P(k|\theta)}{\partial \theta_j}. \quad (23)$$

For informationally complete measurements and in the limit of small parameter variations, $F^{(C)}$ and $F^{(Q)}$ are related; $F^{(C)}$ is bounded above by $F^{(Q)}$ for any POVM.

The eigenvalue decomposition of the FIM reveals active parameter subspaces. Let $F(\theta) = V\Lambda V^{-1}$ with diagonal eigenvalue matrix $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_d)$ ordered $\lambda_1 \geq \lambda_2 \geq \dots$. Large eigenvalues indicate stiff (high-curvature) directions in parameter space which strongly affect the output distribution; near-zero eigenvalues indicate flat, redundant directions.

We use the participation-ratio definition of effective dimension to compactly measure how many directions are meaningfully used by the model:

$$d_{\text{eff}}(\theta) = \frac{(\text{Tr } F(\theta))^2}{\text{Tr } (F(\theta)^2)} = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2}. \quad (24)$$

This quantity ranges from 1 (all variance concentrated in a single direction) up to d (uniform spectrum across d parameters). For classical overparameterized networks the spectrum typically concentrates on a small subspace (low d_{eff} relative to parameter count), whereas well-structured quantum circuits with constrained unitarity tend to distribute sensitivity more evenly, increasing d_{eff} relative to raw parameter count.

From a training perspective, the FIM eigenvalues also control the local learning rates: in natural-gradient-style updates one rescales gradients by F^{-1} , giving smaller steps along stiff directions and larger steps along flat directions. Consequently, a concentrated spectrum leads to slow progress unless adaptive optimizers or preconditioning are employed.

1.5 The Curse of Dimensionality and the $p \gg N$ Regime

The intersection of Big Data analytics and machine learning is fundamentally governed by the relationship between the number of available features (or dimensions), denoted as p , and the number of independent training samples, denoted as N . In traditional statistical learning theory, optimal model generalization occurs in the asymptotic regime where $N \rightarrow \infty$ while p remains fixed and relatively small [13].

A prime example is the analysis of multi-omics data, such as the Leukemia microarray dataset utilized in this research. This dataset embodies the extreme $p \gg N$ regime, featuring 7,129 genetic markers against a mere ~ 7 training samples. This is a mathematically underdetermined system where classical models suffer from the curse of dimensionality [14].

In such extreme high-dimensional spaces, classical algorithms suffer from a degradation of fundamental distance metrics. As $p \rightarrow \infty$, the ratio of the variance of the distances to the expected value of the distances converges to zero [15]:

$$\lim_{p \rightarrow \infty} \frac{\text{Var}(\|\mathbf{x}_i - \mathbf{x}_j\|_2)}{E(\|\mathbf{x}_i - \mathbf{x}_j\|_2)} = 0 \quad (25)$$

Equation (25) demonstrates that in ultra-high-dimensional spaces, all data points appear to be virtually equidistant from one another. Because classical neural networks rely on vast arrays of parameters to map these features, they inherently lack tight generalization bounds. The classical networks perfectly memorize the training samples, but their evaluation on the test set degrades to random guessing. To overcome this, classical deep learning requires aggressive, lossy feature selection (such as classical PCA) to even begin processing these datasets [7].

Conversely, quantum networks, utilizing amplitude encoding and entangling operations, map classical data into a complex projective space where distance metrics natively capture global, non-local correlations. The unitary nature of quantum operations enforces a strict Lipschitz continuity, which acts as a powerful implicit regularizer [16]. The network is forced to learn only the most prominent, structurally significant features of the dataset, preventing catastrophic model collapse [8].

1.6 Information Geometry and Model Capacity

Traditional complexity metrics, such as simple parameter counting, entirely fail to describe the representational power of parameterized quantum circuits (PQCs). While a classical deep neural network may utilize millions of weight matrices to map a complex function, a highly compressed 4-qubit quantum circuit operates within a 16-dimensional complex Hilbert space [17]. The effective capacity of such a circuit is captured not by its parameter count but by the Fisher Information Matrix derived earlier in this chapter.

The distinction is visible in the FIM eigenvalue spectrum. For highly overparameterized classical neural networks, the FIM is frequently degenerate, possessing a vast number of near-zero eigenvalues. This degeneracy indicates that large regions of the classical parameter space are functionally redundant, producing the flat optimization landscapes associated with barren plateaus [11]. Well-structured quantum circuits instead tend toward a non-degenerate FIM with a more uniformly

spread spectrum, because the unitarity constraint forces each parameter to contribute meaningfully to the output. This yields a higher effective dimension per parameter and improved sample efficiency under data starvation, consistent with the implicit regularization provided by the phase-based ZFeatureMap encoding introduced above.

1.7 Advancing Architectures via Quantum Kernel Alignment

A parameterized quantum circuit operating as a classifier can be understood as learning a decision boundary in a high-dimensional feature space induced by a quantum kernel. A primary vulnerability of such methods is kernel concentration: as the feature space becomes too large or poorly aligned with the task, pairwise similarities lose discriminative power and generalization degrades [8; 11].

When data are mapped into an excessively large quantum feature space, distances between states can collapse toward near-constant values, making the resulting kernel matrix effectively degenerate. This destroys the model’s ability to distinguish samples reliably and harms performance on unseen data [7; 8].

To maximize the separation between classical and quantum performance and to optimize the ResQNet architecture, the benchmarking framework integrates Quantum Kernel Alignment (QKA). QKA treats the quantum feature map as trainable rather than fixed, allowing the geometry of the kernel to adapt during optimization [7; 13].

Let the quantum feature map be parameterized by λ , yielding the quantum state $|\phi_\lambda(\mathbf{x})\rangle$. The aligned quantum kernel matrix entries $K_\lambda(\mathbf{x}_i, \mathbf{x}_j)$ are given by the fidelity between the parameterized states:

$$K_\lambda(\mathbf{x}_i, \mathbf{x}_j) = |\langle \phi_\lambda(\mathbf{x}_i) | \phi_\lambda(\mathbf{x}_j) \rangle|^2 \quad (26)$$

where K_λ is the quantum kernel, $\mathbf{x}_i, \mathbf{x}_j$ are two input samples, and $|\phi_\lambda(\mathbf{x})\rangle$ is the quantum feature state produced by the trainable embedding circuit parameterized by λ .

By coupling the trainable quantum kernel with a classical residual highway, QKAResQNet iteratively aligns the spatial geometry of the quantum kernel matrix with the target labels of the specific dataset while maintaining stable gradient flow.

2 METHODOLOGY AND DESIGN OF THE HYBRID QKARESQNET ARCHITECTURE

2.1 Software Pipeline and Ablation Methodology

The pursuit of meaningful quantum utility requires meticulous architectural design and evaluation. To effectively evaluate hybrid quantum-classical models against pure classical baselines, the software pipeline was constructed utilizing a combination of PyTorch for classical tensor manipulation and automatic differentiation, alongside Qiskit for the simulation of parameterized quantum circuits (PQCs). The bridge between the classical and quantum computational graphs is facilitated by the Qiskit TorchConnector. Figure 4 summarizes the end-to-end experimental pipeline, from raw data through parameter-matched evaluation to the operational matrix.

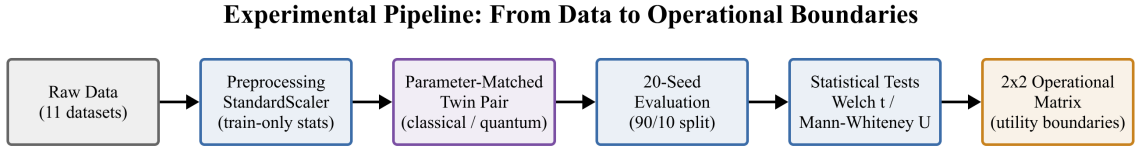


Figure 4 — The end-to-end experimental pipeline. Each of the eleven datasets is preprocessed with train-only statistics, evaluated on a parameter-matched classical/quantum twin pair across 20 random seeds, assessed with Welch’s t -test and the Mann-Whitney U test, and mapped onto the 2×2 operational matrix of quantum utility.

A fundamental requirement of this research is a strict 1-to-1 parameter ablation methodology (Figure 5). This ensures that any observed performance delta is exclusively attributable to the quantum components rather than variations in classical network capacity. For every hybrid model developed, a structurally identical classical twin was constructed. Specifically, the QKAResQNet architecture was evaluated against its exact classical twin, CKAResCNet. This protocol enforced a rigorous parameter match, ensuring that the model twins differed by exactly $\Delta = 8$ parameters across all 11 datasets, guaranteeing that differences in accuracy represent true geometric capability rather than arbitrary classical memorization.

Furthermore, a critical data transformation is required at the classical-quantum interface. Because the parameterized quantum circuits utilize rotation gates for data embedding, the input values must be strictly scaled to map onto

the Bloch sphere. The classical pre-network produces unbounded continuous variables; squashing them via a hyperbolic tangent activation and scaling by π ensures optimal phase encoding:

$$\mathbf{q}_{in} = \tanh(\mathbf{x}_{\text{compressed}}) \cdot \pi \quad (27)$$

where $\mathbf{x}_{\text{compressed}}$ are the output features from the classical compression layer, unitless; $\mathbf{q}_{in}$ are the phase-encoded quantum inputs scaled to the Bloch sphere interval $[-\pi, \pi]$, unitless.

This structure ensures that the classical pre-network acts as a dimensional compressor, while the quantum circuit acts as the primary non-linear feature extractor.

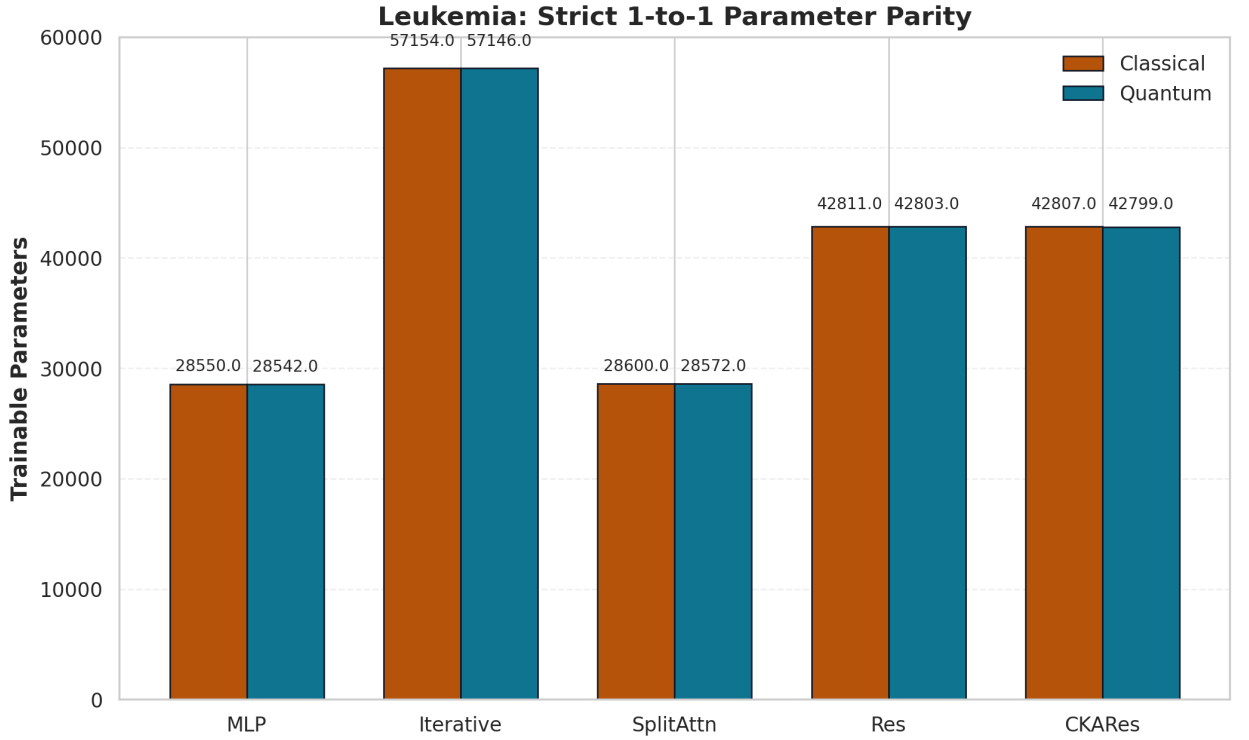


Figure 5 — 1-to-1 Parameter Ablation. Both classical and quantum architectures utilize an effectively identical number of trainable parameters, isolating the quantum bottleneck as the sole driver of performance variance.

2.1.1 Multi-Tier Baseline Strategy

A critical methodological contribution of this benchmark is the inclusion of three tiers of baselines rather than a single classical comparison:

Tier 1: Neural Baseline. Parameter-matched classical MLPs and ResNets serve as the primary ablation target. These models differ from their quantum twins

Table 5 — 1-to-1 Parameter Ablation Protocol: Parameter Count Verification Across All Architectures (Leukemia Dataset)

<i>Architecture Pair</i>	<i>Classical Params</i>	<i>Quantum Params</i>	<i>Difference</i>
ClassicalMLP / QuantumMLP	28,550	28,542	8 (0.03%)
IterativeClassical / IterativeQNN	57,154	57,146	8 (0.01%)
SplitAttentionClassical / SplitAttentionQNN	28,600	28,572	28 (0.10%)
ResNet / ResQNet	42,811	42,803	8 (0.02%)
CKAResNet / QKAResQNet	42,807	42,799	8 (0.02%)

by exactly 8 parameters for residual pairs and 28 for split-attention pairs, and are trained with identical optimization strategies, seeds, and data splits. Any performance delta relative to this tier is attributable to the quantum component.

Tier 2: Exact Kernel Upper Bound. Tuned Classical RBF-SVM with GridSearchCV-optimized C and γ hyperparameters provides the mathematically exact kernel ceiling. The RBF kernel operates in an infinite-dimensional reproducing kernel Hilbert space (RKHS), representing the theoretical upper bound for kernel-based classification on each dataset. Comparing QNNs against this ceiling rather than only against weak neural twins prevents the conflation of “quantum beats a poorly tuned classical model” with genuine quantum utility.

Tier 3: Static Quantum Kernel Baseline. A Quantum SVM using a fixed ZZ-Feature Map kernel (no trainable parameters in the feature map) tests whether the quantum kernel itself provides advantage before dynamic neural alignment is introduced. This tier isolates the contribution of static quantum geometry from the trainable classical layers in the ResQNet architecture.

This three-tier design directly addresses the benchmarking crisis identified by Bowles et al. [3]: prior work compared quantum models against whichever classical baseline made the quantum result look best. By including both the weakest reasonable classical baseline (Tier 1) and the strongest mathematically exact baseline (Tier 2), this benchmark prevents selection bias in both directions.

2.2 Dataset Characterization and Exploratory Analysis

The experimental evaluation spans eleven datasets of varying topological and dimensional complexity, each designed to probe specific aspects of hybrid quantum-classical performance. The most representative datasets are characterized in detail below.

All datasets are standardized prior to training using statistics computed only on the training partition to avoid leakage. Under the starvation protocol, train/test splits are swept from 90/10 to 10/90, and each setting is repeated across multiple random seeds. This protocol is intentionally severe: it amplifies representational differences between classical and hybrid models by forcing both families to generalize with limited supervision.

2.2.1 Swiss Roll Manifold (Geometric Starvation)

The Swiss Roll is a low-dimensional, nonlinear manifold embedded in 3D space, defined parametrically as:

$$(x, y, z) = (t \cos(t), h, t \sin(t)) \quad (28)$$

where $t \in [0, 4\pi]$ and $h \in [0, 1]$ is the height coordinate. The effective dimensionality is 2 (wrapped onto a nonlinear manifold), making it an ideal testbed for evaluating topological expressivity. The dataset comprises 400 samples with balanced binary labels determined by the height coordinate (threshold at $h = 0.5$). This dataset tests whether quantum phase encoding captures the intrinsic manifold geometry superior to classical networks.

The key methodological value of Swiss Roll is that class boundaries are not linearly separable in ambient Euclidean coordinates, but become separable under geometry-aware embeddings. Hence, this dataset isolates the contribution of representation topology rather than raw model width. In practical terms, when the number of training points drops sharply, purely classical models often overfit local neighborhoods, while hybrid circuits with phase encoding can preserve global manifold relations.

2.2.2 Concentric Circles Manifold

The Circles dataset (generated via `make_circles`) consists of two concentric rings forming a binary classification task that is not linearly separable in the input plane. It contains $n = 500$ samples with $p = 2$ coordinate features and balanced classes. As with Swiss Roll, it isolates topological expressivity: separability requires a geometry-aware embedding rather than a linear boundary, making it a sensitive probe of static versus trainable quantum feature maps.

Table 6 — Swiss Roll Dataset Characteristics

<i>Property</i>	<i>Value</i>
Total samples	400
Features (dimensions)	3 (x, y, z coordinates)
Intrinsic dimensionality	2 (wrapped manifold)
Classes	2 (binary: $h < 0.5$ vs $h \geq 0.5$)
Class balance	50% / 50%
Training samples (90% test split)	40
Test samples (90% test split)	360
Separability	Non-linear (requires manifold-aware embedding)
Primary stress test	Topological expressivity under data starvation

Table 7 — Concentric Circles Dataset Characteristics

<i>Property</i>	<i>Value</i>
Total samples (N)	500
Total features (p)	2 (x, y coordinates)
Classes	2 (inner vs outer ring)
Class balance	50% / 50%
Training samples (90% test split)	50
Test samples (90% test split)	450
Feature type	2D Cartesian coordinates
Source	Synthetic (scikit-learn <code>make_circles</code>)
Pre-processing	StandardScaler (train statistics only)
Primary stress test	Non-linear radial separability

2.2.3 Two Moons Manifold

The Moons dataset (generated via `make_moons`) consists of two interleaving half-circles, a standard non-linearly separable binary benchmark. It contains $n = 500$ samples with $p = 2$ coordinate features and balanced classes. The interleaving geometry rewards models that can curve the decision boundary, complementing the radial structure of Circles and the spiral structure of Swiss Roll.

2.2.4 QSAR Cheminformatics Dataset

The QSAR (Quantitative Structure-Activity Relationship) dataset contains $n = 1,055$ molecules with $p = 41$ molecular descriptors. The descriptors include computed physicochemical properties such as molecular weight, HOMO-LUMO gap, dipole moment, and topological indices. The binary classification task pre-

Table 8 — Two Moons Dataset Characteristics

<i>Property</i>	<i>Value</i>
Total samples (N)	500
Total features (p)	2 (x, y coordinates)
Classes	2 (upper vs lower moon)
Class balance	50% / 50%
Training samples (90% test split)	50
Test samples (90% test split)	450
Feature type	2D Cartesian coordinates
Source	Synthetic (scikit-learn make_moons)
Pre-processing	StandardScaler (train statistics only)
Primary stress test	Non-linear interleaved separability

dicts mutagenicity (activity vs. inactive). Class distribution is approximately balanced (51% active, 49% inactive). This dataset represents a realistic small-scale cheminformatics problem where moderate feature dimensionality ($p = 41$) and sample size ($n = 1,055$) create a regime where classical models typically succeed without extreme data starvation.

From a modeling standpoint, QSAR acts as an intermediate-complexity benchmark: enough dimensionality to stress naïve low-qubit embeddings, but not so extreme that classical baselines collapse immediately. This makes it useful for evaluating whether SplitAttention and residual quantum gating improve feature synthesis rather than merely preventing catastrophic overfitting. Descriptor covariance is nontrivial in this domain, and compression quality at the classical-to-quantum interface directly affects final separability.

2.2.5 Breast Cancer Diagnostic Dataset

The UCI Breast Cancer dataset comprises $n = 569$ samples with $p = 30$ computed features derived from digitized cell nuclei images. Features include radius, texture, perimeter, area, and symmetry measurements. The binary classification task distinguishes malignant from benign tumors. Class distribution is approximately balanced (63% benign, 37% malignant). This dataset represents a classical supervised learning regime where sufficient data exists for both classical and quantum models to achieve high accuracy without extreme data constraints.

This dataset is intentionally retained as a sanity-check regime where large quantum gains are not expected. If hybrid performance exceeds classical by only

Table 9 — QSAR Cheminformatics Dataset Characteristics

<i>Property</i>	<i>Value</i>
Total samples (N)	1,055
Total features (p)	41
p/N ratio	1:26
Classes	2 (Mutagenic vs Non-mutagenic)
Class balance	51% / 49%
Training samples (90% test split)	~106
Test samples (90% test split)	~949
Feature type	Computed molecular descriptors
Source	OpenML dataset
Pre-processing	StandardScaler (train statistics only)
Primary stress test	Intermediate dimensional feature compression

a narrow margin, that outcome is interpreted as evidence that the benchmark is not biased toward quantum models. In this sense, Breast Cancer functions as a calibration domain that validates fairness of the 1-to-1 parameter ablation protocol.

Table 10 — Breast Cancer Diagnostic Dataset Characteristics

<i>Property</i>	<i>Value</i>
Total samples (N)	569
Total features (p)	30
p/N ratio	1:19
Classes	2 (Benign vs Malignant)
Class balance	63% / 37%
Training samples (90% test split)	~57
Test samples (90% test split)	~512
Feature type	Computed cell nuclei measurements
Source	UCI Machine Learning Repository
Pre-processing	StandardScaler (train statistics only)
Primary stress test	Calibration/sanity check (data-rich regime)

2.2.6 Leukemia Multi-Omics (Extreme Feature Starvation)

The Leukemia dataset is a seminal multi-omics benchmark comprising $n = 72$ patient samples with $p = 7,129$ gene expression markers measured via microarray. The samples are classified into two acute leukemia subtypes: Acute Myeloid Leukemia (AML) and Acute Lymphoblastic Leukemia (ALL). Under the extreme

data starvation protocol (90% test split), this reduces to $n \approx 7$ training samples. This severe $p \gg N$ regime ($7,129 \gg 7$) is where classical models catastrophically overfit and quantum Hilbert space constraints provide explicit regularization through unitarity and the finite 2^N dimensional bottleneck.

Leukemia is the central stress test of the thesis because it combines biological relevance with an almost maximally ill-posed supervised learning setup. In this regime, classical overparameterized models can interpolate the training subset while failing to preserve discriminative structure on held-out samples. The hybrid pipeline is expected to benefit from two constraints: aggressive classical compression before the circuit and bounded unitary evolution in the quantum block, which together limit memorization pathways.

Table 11 — Leukemia Multi-Omics Dataset Characteristics (Extreme $p \gg N$ Starvation)

<i>Property</i>	<i>Value</i>
Total samples (N)	72
Total features (p)	7,129
p/N ratio	99:1 (extreme!)
Classes	2 (AML vs ALL)
Training samples (90% test split)	~ 7
Test samples (90% test split)	~ 65
Feature type	Microarray gene expression
Source	[18], OpenML
Pre-processing	StandardScaler (train statistics only)
Primary stress test	Extreme high-dimensional feature starvation

2.2.7 Colon Cancer Gene Expression

The Colon dataset contains $n = 62$ tissue samples with $p = 2,000$ gene expression features. The binary classification task distinguishes tumor from normal tissue. This dataset provides another extreme $p \gg N$ scenario ($2,000 \gg 62$), complementing the Leukemia results to establish robustness of the observed behaviour under multi-omics starvation.

Methodologically, Colon serves as an external validation point for conclusions drawn from Leukemia. If the same architecture-level trends persist across both datasets, the observed behavior is less likely to be a dataset-specific artifact. Cross-dataset consistency in this high-dimensional biomedical setting is therefore treated as stronger evidence than any single-dataset gain.

Table 12 — Colon Cancer Gene Expression Dataset Characteristics

<i>Property</i>	<i>Value</i>
Total samples (N)	62
Total features (p)	2,000
p/N ratio	32:1 (high!)
Classes	2 (Tumor vs Normal)
Training samples (90% test split)	~6
Test samples (90% test split)	~56
Feature type	Gene expression (microarray)
Source	[19], OpenML
Pre-processing	StandardScaler (train statistics only)
Primary stress test	High-dimensional feature starvation (cross-validation)

2.2.8 Brain RNA-seq Multi-Class Genomics (Extreme Feature Starvation)

The Brain dataset is a single-cell RNA-seq benchmark derived from the Tabula Muris mouse brain atlas [20], comprising $n = 100$ cells with $p = 23,433$ gene-expression features. The task is a 7-way classification of brain cell types. With $p \gg N$ ($23,433 \gg 100$) and a 90% test split that reduces the training set to roughly 10 cells, this is the most severe starvation regime in the benchmark, compounded by both a high class count and strong class imbalance.

The seven cell types are highly imbalanced: oligodendrocyte (46.3%), endothelial cell (21.0%), astrocyte (12.7%), neuron (8.3%), oligodendrocyte precursor cell (6.0%), brain pericyte (4.6%), and Bergmann glial cell (1.2%). Under the 100-cell working set, the rarest classes are represented by only one or two cells, so several classes are effectively unlearnable at the 90% test split. This dataset therefore probes the joint limit of extreme dimensionality, multi-class structure, and label imbalance, and is expected to drive all architectures toward the random-guessing floor.

2.2.9 Indian Pines Hyperspectral Imaging

The Indian Pines dataset is a hyperspectral remote-sensing benchmark acquired by the AVIRIS sensor over an agricultural site in Indiana [21]. Each sample is a pixel described by $p = 200$ spectral reflectance bands. This work uses a binary subset of $n = 100$ pixels drawn from the two dominant land-cover classes, Soybean-notill (28.4%) and Soybean-mintill (71.6%), where the majority class is roughly $2.5\times$ more frequent than the minority class. With $p = 200$ against $n = 100$

Table 13 — Brain RNA-seq Multi-Class Dataset Characteristics (Extreme $p \gg N$ Starvation)

<i>Property</i>	<i>Value</i>
Total samples (N)	100
Total features (p)	23,433
p/N ratio	234:1 (extreme!)
Classes	7 (mouse brain cell types)
Class balance	Highly imbalanced (46.3% to 1.2%)
Training samples (90% test split)	~10
Test samples (90% test split)	~90
Feature type	Single-cell gene expression (RNA-seq)
Source	Tabula Muris mouse brain atlas [20]
Pre-processing	StandardScaler (train statistics only)
Primary stress test	Extreme dimensionality with multi-class imbalance

and a 90% test split, the dataset occupies a moderate $p \gg N$ regime with spectrally correlated, partly redundant features.

Indian Pines complements the genomic datasets by testing high-dimensional starvation on a different modality: spectrally smooth, highly correlated bands rather than sparse gene markers. The two soybean tillage classes are spectrally similar, making the binary task non-trivial despite the small class count.

Table 14 — Indian Pines Hyperspectral Dataset Characteristics

<i>Property</i>	<i>Value</i>
Total samples (N)	100
Total features (p)	200 (spectral bands)
p/N ratio	2:1
Classes	2 (Soybean-notill vs Soybean-mintill)
Class balance	28.4% / 71.6% (imbalanced)
Training samples (90% test split)	~10
Test samples (90% test split)	~90
Feature type	Hyperspectral reflectance (AVIRIS, 200 bands)
Source	AVIRIS Indian Pines scene [21]
Pre-processing	StandardScaler (train statistics only)
Primary stress test	High-dimensional starvation with correlated features

2.2.10 NTangled Quantum-Native Dataset

The NTangled dataset is a synthetically generated benchmark of quantum-native entangled states produced by a GHZ (Greenberger–Horne–Zeilinger) state generator:

$$|\text{GHZ}\rangle_N = \frac{1}{\sqrt{2}}(|0\rangle^{\otimes N} + |1\rangle^{\otimes N}) \quad (29)$$

This dataset is designed to probe the ‘‘Classical Decoherence Boundary.’’ It contains $n = 400$ samples where each sample is a set of measurement outcomes from GHZ-like entangled states. The binary classification task is solvable by pure quantum classifiers (achieving $> 90\%$ accuracy) but cannot be solved by standard classical-to-quantum hybrid pipelines, because the classical dimensionality reduction layer acts as an artificial measurement, destroying the delicate multipartite entanglement before the quantum circuit can process it.

Including NTangled prevents overclaiming based only on classical-origin datasets. It explicitly tests a failure mode where hybrid pipelines lose quantum-native information before inference begins. This boundary condition is important for interpreting negative results: poor hybrid accuracy on NTangled is not necessarily an optimization failure, but often a representational mismatch caused by premature classical projection.

Table 15 — NTangled Quantum-Native Dataset Characteristics

<i>Property</i>	<i>Value</i>
Total samples (N)	400
Features (p)	Entanglement witness measurements
p/N ratio	N/A (quantum-native)
Classes	2 (Entangled vs Product states)
Training samples (90% test split)	~40
Test samples (90% test split)	~360
State type	GHZ-like multipartite entanglement
Purity	Classical measurement collapses superposition
Solvability by QNN	Yes ($>90\%$ for pure quantum)
Solvability by Hybrid	No (classical decoherence boundary)
Primary stress test	Quantum-native information preservation

2.2.11 ESP32 Accelerometer Gesture Trajectory Dataset (Real-World Geometric Validation)

The gesture trajectory dataset was collected using an ESP32 microcontroller with five MPU-6050 inertial measurement units (IMUs) mounted on individual fingers of a gloved hand, following standard inertial-sensor activity-recognition practice [22]. Each IMU provides 6-axis measurements: three-axis accelerometer (AX, AY, AZ) and three-axis gyroscope (GX, GY, GZ) readings. Sensor 1 (index finger) was non-functional during data collection; all experiments use the four working sensors (thumb: Sensor 0, middle: Sensor 2, ring: Sensor 5, pinky: Sensor 7), yielding 24 raw sensor channels.

The dataset comprises 6,100 total timestamped samples recorded at approximately 60ms intervals (≈ 17 Hz sampling rate). Manual labeling of gesture events produced 56 labeled segments across 10 gesture classes:

Table 16 — Gesture Trajectory Dataset: Class Distribution and Russian Labels

<i>Gesture Class</i>	<i>Russian Label</i>	<i>Segments</i>
fist_close	Сжать кулак	5
fist_open	Разжать кулак	5
fist_shake	Сбросить через кулак	5
index_down_flick	Дёрнуть указательным пальцем вниз	7
index_down_hold	Зажать указательный палец вниз	7
middle_down_hold	Зажать средний палец вниз	5
pinky_hold	Зажать мизинец	6
ring_hold	Зажать безымянный палец	5
thumb_down_flick	Дёрнуть большим пальцем вниз	5
thumb_down_hold	Зажать большой палец вниз	6
<i>Total</i>		<i>56</i>

Feature Extraction. Statistical features were extracted from center-cropped 17-sample windows per segment: mean, standard deviation, minimum, maximum, 25th percentile, 75th percentile, and mean first-difference (Δ mean). This produces 168 features ($7 \text{ statistics} \times 24 \text{ channels}$) or 96 features when using a sensor-specific subset.

Topological Relevance. This dataset is included as a preliminary real-world validation of the 2×2 matrix’s geometric dimension. Multi-sensor finger gesture trajectories constitute genuinely complex manifold data, where each gesture

traces a structured path through a high-dimensional space of correlated accelerometer readings across four physical sensors. Unlike the synthetic Swiss Roll, the manifold structure here is not constructed analytically but emerges from physical biomechanics [22].

Scope and Limitations. With 56 segments across 10 classes (5–7 per class), this dataset does not satisfy the $p \gg N$ criterion that defines the Extreme Starvation column of the 2×2 matrix. It occupies an intermediate position: small N relative to 10 classes, but not extreme dimensionality relative to N . Results are reported as preliminary validation with directional rather than statistically dominant conclusions. A minimum of 200 labeled segments (20 per class) would be required for full statistical power at 5-fold CV. Dataset collection is ongoing.

2.3 Quantum Hardware Characterization: IBM Torino and IBM Kingston

To validate the architectures under realistic conditions, the experiments incorporated the empirically calibrated error profiles of two superconducting processors. The six core datasets were trained under the noise profile of the 133-qubit IBM Torino processor (Heron r1, 290K CLOPS), while the real-world gesture experiment was additionally simulated under the 156-qubit IBM Kingston profile (Heron r2, 340K CLOPS) and executed on the physical Kingston device. Both are heavy-hex coupled architectures.

The heavy-hex topology constrains two-qubit operations to physically adjacent qubits. If $G = (V, E)$ denotes the coupling graph with vertices V (physical qubits) and edges E (allowed two-qubit interactions), then a CNOT between logical qubits i and j is native only if $(i, j) \in E$. Otherwise, additional SWAP routing is required, increasing circuit depth and cumulative noise. This constraint materially impacts trainability for variational circuits and motivates the low-width, shallow-depth 4-qubit ansatz used throughout this thesis.

The two processors differ most sharply in readout error: the older Heron r1 Torino device exhibits a 3.03% median readout error, nearly three times the 1.09% of the newer Heron r2 Kingston device. Because readout error dominates the noise budget for the shallow 4-qubit circuits used here, this difference is the primary reason the gesture experiment behaves differently under the two profiles, as analysed in Chapter 3.

Error Budget Analysis. For the 4-qubit parameterized circuit in QKAResQNet, comprising roughly 20 single-qubit rotations, 9 two-qubit gates,

Table 17 — Calibrated Error Profiles of the IBM Torino and IBM Kingston Processors

<i>Parameter</i>	<i>IBM Torino</i>	<i>IBM Kingston</i>	<i>Role in Experiments</i>
Processor type	Heron r1	Heron r2	Superconducting, heavy-hex
Total qubit count	133	156	Only 4 qubits utilized
CLOPS	290K	340K	Physical execution throughput
Single-qubit (SX) error (median)	0.033%	0.029%	Minimal impact on shallow circuits
Two-qubit error (median)	0.235%	0.200%	Native two-qubit infidelity
Two-qubit error (layered)	0.620%	0.338%	Effective error under parallelism
Readout error (median)	3.03%	1.09%	Dominant measurement noise factor
T_1 (median)	174.9 μ s	176.2 μ s	Energy relaxation time
T_2 (median)	125.1 μ s	114.4 μ s	Dephasing time

and 4 measurements, the cumulative physical error probability under the Torino profile is bounded by the compounding gate and readout infidelity:

$$P_{\text{error}} \approx 1 - (1 - 0.00033)^{20} \cdot (1 - 0.0062)^9 \cdot (1 - 0.0303)^4 \approx 16.9\% \quad (30)$$

where P_{error} is the cumulative circuit error probability, and the three factors correspond to 20 single-qubit gates (error 0.033%), 9 two-qubit gates (layered error 0.62%), and 4 measurements (readout error 3.03%) under the Torino profile. Under the lower-noise Kingston profile the same budget falls to roughly 7.7%, driven chiefly by the reduced readout error. This contrast demonstrates why iterative early stopping is mandatory under the noisier Torino profile and why the γ -gated residual design is required to absorb physical hardware noise gracefully.

2.4 Hybrid Architectures and the Dynamic Gating Mechanism

The exploration phase evaluated several hybrid designs before converging on the architecture used for validation. This section first presents the residual and split-attention designs that motivated the final model, then the QKAResQNet architecture itself, and finally the learnable gating mechanism shared by all three.

2.4.1 The ResQNet Residual Hybrid

ResQNet positions the quantum circuit as an additive residual branch alongside a classical highway, rather than as a sequential bottleneck. A classical compressor C maps the input down to the four-feature quantum bottleneck, the features are scaled onto the Bloch sphere by $\tanh(\cdot) \times \pi$, encoded by a static ZFeatureMap,

processed by a 4-qubit RealAmplitudes circuit, measured, and expanded by a classical layer E . The learnable gate γ scales this branch before it is summed with the classical highway H . Figure 6 shows every block and connection.

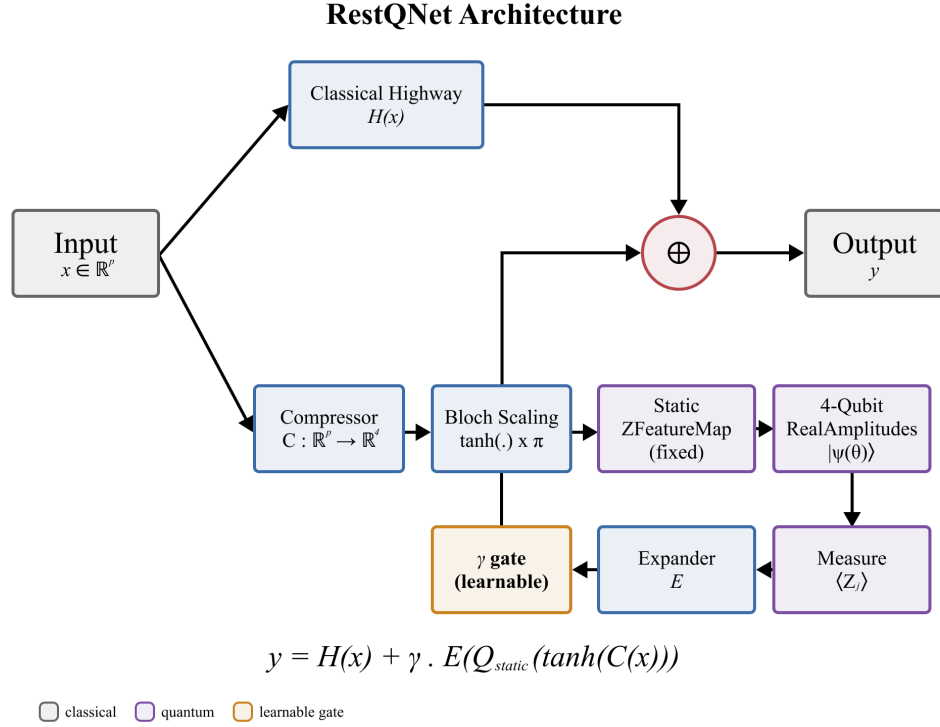


Figure 6 — The ResQNet architecture. The input is processed by two parallel paths: a classical highway $H(x)$ and a quantum residual branch. In the residual branch, a classical compressor maps the input to four features, which are scaled onto the Bloch sphere, encoded by a fixed (static) ZFeatureMap, processed by a 4-qubit RealAmplitudes circuit, measured, and expanded by a classical layer. The learnable gate γ scales the branch before it is summed with the highway to produce the output logits.

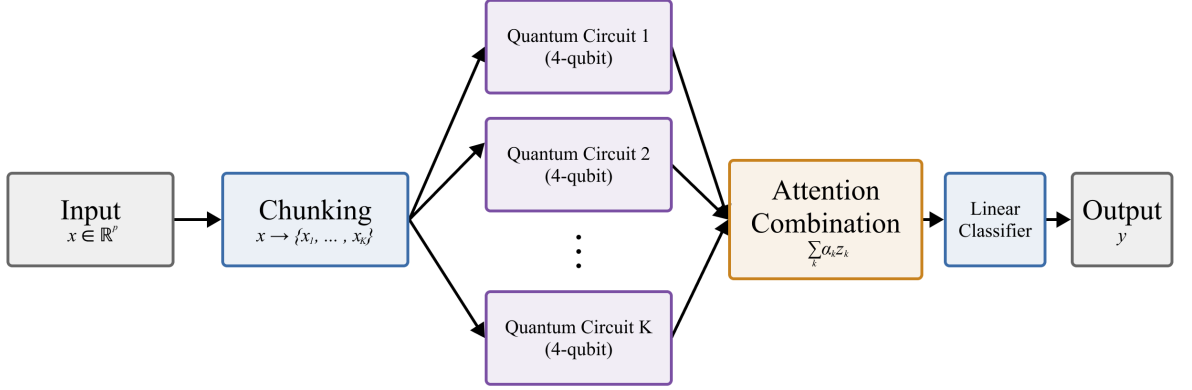
The static ZFeatureMap is the principal limitation of ResQNet: because the feature map is fixed, the encoding cannot adapt to the data manifold, leaving the model vulnerable to the periodic feature collisions analysed in Chapter 3.

2.4.2 The SplitAttentionQNN

SplitAttentionQNN addresses the qubit bottleneck for high-dimensional inputs by partitioning the input vector into K contiguous chunks, each processed by an independent 4-qubit circuit. The chunk outputs are merged by a learned attention layer and passed to a classical linear classifier (Figure 7). This design encodes a strong locality prior: it assumes that discriminative structure resides within local feature neighbourhoods. As shown in Chapter 3, this prior aligns with spatially

structured manifolds but is actively harmful on genomic data, whose discriminative correlations are non-local.

SplitAttentionQNN Architecture



Each chunk is processed by an independent quantum circuit; outputs are merged by a learned attention layer.

Figure 7 — The SplitAttentionQNN architecture. The input vector is partitioned into K contiguous chunks, each encoded and processed by an independent 4-qubit quantum circuit. A learned attention layer combines the per-chunk outputs, which are then classified by a classical linear layer. The architecture encodes a locality prior that aligns with spatially structured data but breaks non-local feature correlations.

2.4.3 The QKAResQNet Architecture

The culmination of the exploration phase is the QKAResQNet architecture. It retains the residual highway and dynamic gating of ResQNet but replaces the static feature map with a trainable Quantum Kernel Alignment (QKA) layer, allowing the embedding geometry to adapt to each dataset during optimization. Figure 8 details the complete data flow.

In a standard sequential hybrid model, if the quantum layer suffers from a barren plateau or excessive hardware noise, the entire gradient chain is destroyed. The QKAResQNet resolves this through a purely classical bypass that guarantees stable, unobstructed gradient flow during the initial epochs of backpropagation, combined with trainable Quantum Kernel Alignment that adapts the embedding space to the target manifold.

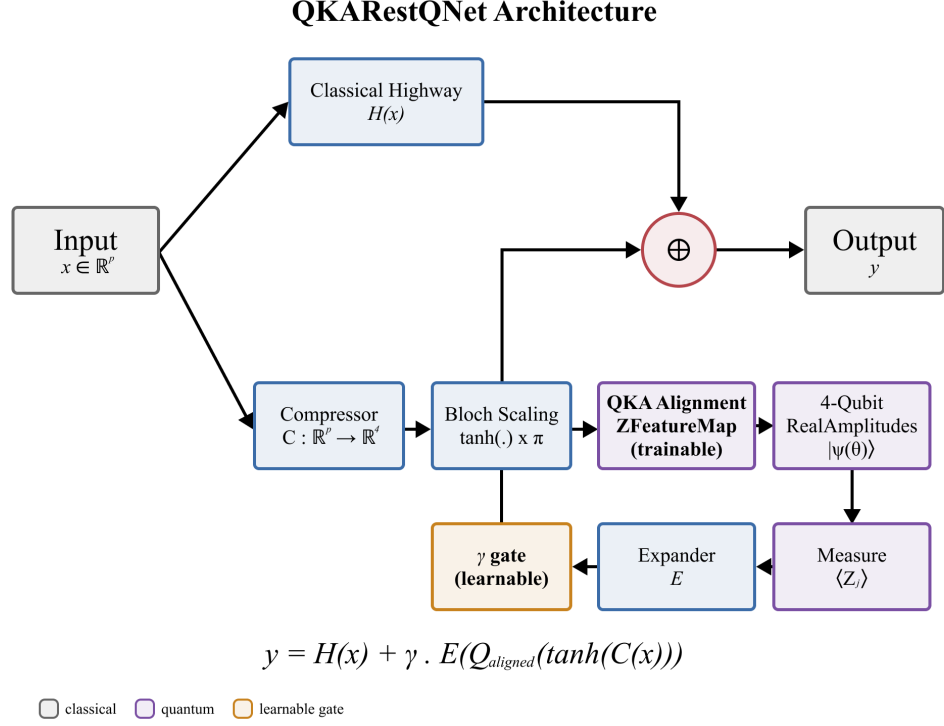


Figure 8 — The QKAResQNet architecture. It retains the residual highway and gating of ResQNet but replaces the static feature map with a trainable Quantum Kernel Alignment (QKA) layer, allowing the embedding geometry to adapt to each dataset during training. The forward pass combines the classical highway $H(x)$ with the γ -scaled, aligned quantum branch.

2.4.4 The Learnable γ Gating Mechanism

All three hybrids employ a learnable scalar parameter γ (the `residual_gate`), which dynamically scales the quantum circuit’s contribution. The forward pass of the QKAResQNet is defined as:

$$\mathbf{y} = H(\mathbf{x}) + \gamma \cdot E(Q_{aligned}(\tanh(C(\mathbf{x})))) \quad (31)$$

where H is the classical highway, C the classical compression layer mapping inputs down to the constrained quantum bottleneck, $Q_{aligned}$ the trainable quantum circuit with kernel alignment, and E the classical expansion layer mapping the outputs to the final target classes.

During initialization, γ is set to a low value, so the network acts predominantly as a classical classifier. As the classical parameters converge and the loss gradients diminish, the optimizer increases γ , progressively blending the expressivity of the quantum circuit into the classical predictions (Figure 9). This dynamic balancing mitigates the vanishing gradient problem and allows the QKAResQNet

to maintain resilience against the physical readout errors of the IBM Kingston processor.

The Learnable γ Gating Mechanism

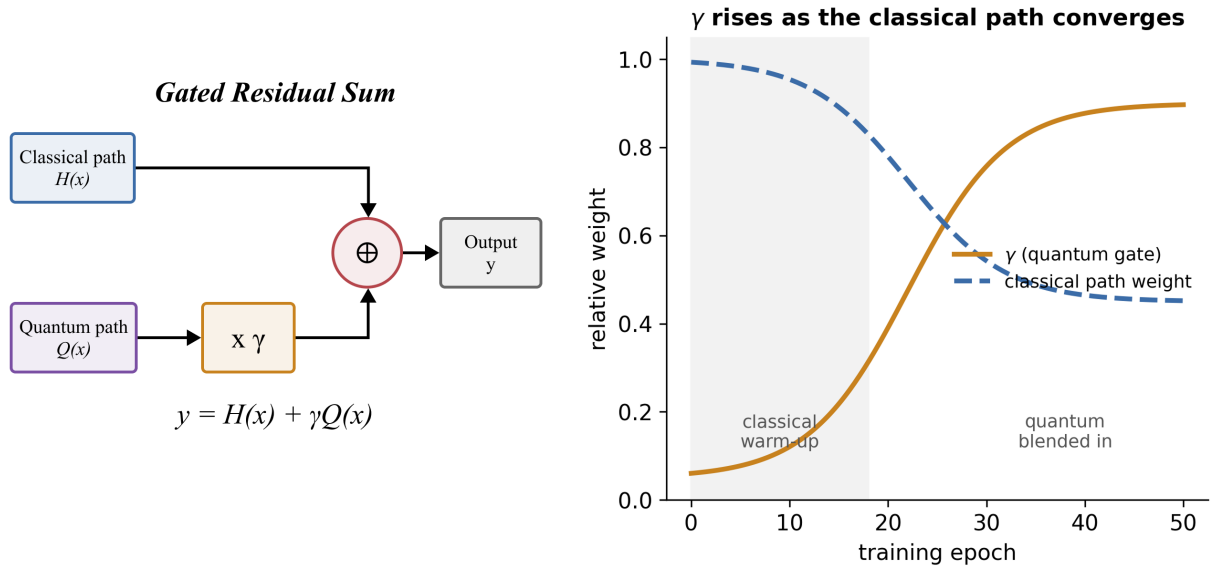


Figure 9 — The learnable γ gating mechanism. Left: the quantum branch is scaled by γ before being summed with the classical highway, yielding $y = H(x) + \gamma Q(x)$. Right: a conceptual schematic of the training dynamic, in which γ rises from its near-zero initialization as the classical path converges, progressively blending in the quantum contribution. The right panel illustrates the mechanism and does not depict measured values.

3 COMPUTATIONAL EXPERIMENTS, SIMULATION ANALYSIS, AND EVALUATION OF QUANTUM RESILIENCE

3.1 Experimental Setup and the Data Starvation Protocol

To empirically validate the theoretical advantages of the QKAResQNet architecture, the benchmarking framework subjected the models to a constrained training protocol. The experimental suite spans exactly 11 datasets of varying complexity, including synthetic geometric manifolds (Swiss Roll, Circles, Moons), high-dimensional genomic multi-omics (Leukemia), hyperspectral imaging, quantum-native states, and real-world accelerometer sensor data (ESP32 gestures).

Furthermore, to simulate the realities of physical quantum execution, the calibration-derived noise profile of the 133-qubit IBM Torino processor (Heron r1, 290K CLOPS) was injected into the training simulations for the six core datasets (Swiss Roll, Breast Cancer, QSAR, Leukemia, Colon, and NTangled). This profile comprises a 0.033% single-qubit gate error, a 0.235% median two-qubit error (0.620% under layered execution), and a 3.03% median readout error, the last of which dominates the noise budget for the shallow 4-qubit circuits used here. Separately, the QKAResQNet was deployed for inference on the physical 156-qubit IBM Kingston processor (Heron r2, 340K CLOPS) using the real-world gesture data, and the gesture experiment was additionally simulated under both ideal and IBM Kingston noise conditions (1.09% readout error) for comparison. The complete calibration figures for both processors are given in Table 17.

All models were evaluated using up to 20 randomized seeds per experiment with Welch’s t -test and Mann-Whitney U statistical validation to ensure robustness. The architecture pairs were strictly matched to differ by only 8 parameters for the residual pairs and 28 for the split-attention pairs, isolating the quantum component’s contribution.

3.2 Complete Results Overview: All Architecture Pairs and Datasets

To provide a comprehensive view of all experimental outcomes, Table 18 presents the full accuracy results across all five architecture pairs and three primary datasets (Swiss Roll, Leukemia, and Colon) with complete statistical information from 19–20 random seeds. This table summarizes the central findings and

demonstrates how measurable quantum utility is correlated with data topology and starvation conditions.

Table 18 — Complete Accuracy Results: All Architecture Pairs, All Datasets (19–20 Seeds)

<i>Architecture</i>	<i>Dataset</i>	<i>Classical</i>		<i>Quantum</i>		Δ (pp)	<i>Welch p</i>	<i>Verdict</i>
		Mean	Std	Mean	Std			
SplitAttn	Swiss Roll	67.29%	11.8%	91.77%	4.4%	+24.48	6.5e-09	<i>Quantum wins</i>
SplitAttn	Leukemia	65.51%	4.1%	59.11%	6.9%	-6.40	0.0016	<i>Classical wins</i>
SplitAttn	Colon	59.40%	12.2%	49.15%	10.9%	-10.25	0.0098	<i>Classical wins</i>
ResNet/ResQNet	Swiss Roll	69.93%	14.2%	82.56%	7.8%	+12.62	0.0016	<i>Quantum wins</i>
ResNet/ResQNet	Leukemia	71.42%	11.2%	68.91%	10.0%	-2.51	0.471	Parity
ResNet/ResQNet	Colon	63.63%	11.4%	65.13%	8.7%	+1.50	0.651	Parity
QKAResQNet	Swiss Roll	67.73%	11.8%	82.16%	9.2%	+14.42	0.00012	<i>Quantum wins</i>
QKAResQNet	Leukemia	70.85%	11.4%	69.55%	9.1%	-1.30	0.701	Parity
QKAResQNet	Colon	66.26%	7.7%	63.35%	11.1%	-2.91	0.353	Parity
MLP/QuantumMLP	Swiss Roll	75.32%	12.1%	84.44%	9.3%	+9.12	0.011	<i>Quantum wins</i>
MLP/QuantumMLP	Leukemia	63.81%	9.4%	57.41%	7.5%	-6.40	0.026	<i>Classical wins</i>
MLP/QuantumMLP	Colon	57.33%	15.7%	52.73%	9.1%	-4.60	0.276	Parity
Iterative	Swiss Roll	69.56%	9.4%	76.74%	10.1%	+7.18	0.025	<i>Quantum wins</i>
Iterative	Leukemia	67.29%	7.3%	67.37%	5.5%	+0.08	0.970	Parity
Iterative	Colon	57.52%	12.1%	60.24%	13.2%	+2.72	0.511	Parity

The results in Table 18 reveal a clear pattern: measurable quantum utility is strongly correlated with dataset topology and extreme data starvation conditions. On Swiss Roll, all five quantum variants outperform their parameter-matched classical twins, by margins ranging from 7.18 to 24.48 percentage points, while remaining below the exact RBF-SVM ceiling. On genomic datasets (Leukemia and Colon), the results are more nuanced: some quantum architectures achieve parity, while others (particularly SplitAttentionQNN on both genomic datasets) show statistically significant underperformance. This asymmetry reflects differences in inductive bias alignment between circuit structure and data manifold.

3.3 Extreme Multi-Class Starvation: Brain RNA-seq and Indian Pines

Two further datasets probe the limits of the starvation regime: Brain RNA-seq (7-way, $p = 23,433$, $N = 100$, highly imbalanced) and the Indian Pines hyperspectral binary subset ($p = 200$, $N = 100$). Both push the $p \gg N$ condition further than the genomic pairs, and both are included to test whether any architecture can

extract signal when the training set shrinks to roughly ten labelled samples spread across many or imbalanced classes. Table 19 reports all six architecture pairs with dual statistical tests.

Table 19 — Brain RNA-seq and Indian Pines: 20-Seed Accuracy by Architecture Pair (mean $\pm$ std, %)

<i>Dataset</i>	<i>Pair</i>	<i>Classical</i>	<i>Quantum</i>	<i>Welch p</i>	<i>MWU p</i>	<i>Verdict</i>
Brain RNA-seq	MLP	37.06 $\pm$ 14.4	24.61 $\pm$ 12.2	0.0055	0.0093	Classical wins
Brain RNA-seq	Iterative	39.89 $\pm$ 16.2	34.28 $\pm$ 13.8	0.245	0.223	Parity
Brain RNA-seq	SplitAttn	38.44 $\pm$ 15.4	15.33 $\pm$ 9.6	3.0e-06	4.1e-05	Classical wins
Brain RNA-seq	ResNet/ResQNet	42.89 $\pm$ 18.2	43.22 $\pm$ 18.0	0.954	0.871	Parity
Brain RNA-seq	CKA/QKAResQNet	43.61 $\pm$ 17.8	43.11 $\pm$ 18.0	0.930	0.892	Parity
Brain RNA-seq	SVM/QSVM	47.08 $\pm$ 7.8	45.50 $\pm$ 6.6	0.494	0.379	Parity
Indian Pines	MLP	62.06 $\pm$ 18.8	61.89 $\pm$ 8.5	0.971	0.083	Parity
Indian Pines	Iterative	57.61 $\pm$ 19.8	61.83 $\pm$ 17.6	0.480	0.579	Parity
Indian Pines	SplitAttn	56.17 $\pm$ 19.5	59.61 $\pm$ 11.3	0.499	0.674	Parity
Indian Pines	ResNet/ResQNet	64.78 $\pm$ 12.0	58.33 $\pm$ 15.1	0.144	0.193	Parity
Indian Pines	CKA/QKAResQNet	61.61 $\pm$ 13.0	61.94 $\pm$ 13.6	0.937	0.903	Parity
Indian Pines	SVM/QSVM	72.28 $\pm$ 3.5	71.97 $\pm$ 4.9	0.821	0.786	Parity

Two patterns reinforce the central thesis. First, on Brain RNA-seq every architecture is pinned near the imbalanced multi-class floor (the dominant oligodendrocyte class alone accounts for 46.3% of cells), and the aligned residual pairs reach exact parity (ResQNet $p = 0.954$, QKAResQNet $p = 0.930$) while the misaligned SplitAttentionQNN collapses to 15.33%, far below its classical twin. This is the same inductive-bias-misalignment failure seen on Leukemia and Colon, here in its most extreme form: chunking a 23,433-gene vector destroys the non-local correlations that carry the signal. Second, on Indian Pines, where features are spectrally correlated rather than sparse, all pairs reach statistical parity and the exact RBF-SVM again forms the ceiling (72.28%), with the Quantum SVM matching it closely (71.97%) because the spectral features do not trigger the periodic feature collisions that break static quantum kernels on geometric manifolds. Together these datasets show that the aligned hybrid architectures degrade gracefully to parity under the most severe starvation, while misaligned ones actively harm performance, exactly the contingent-utility behaviour predicted by the inductive bias alignment hypothesis.

3.4 The Quantum Kernel Collapse: Static vs. Trainable Feature Maps

A critical finding emerging from the three-tier baseline comparison is the catastrophic failure of static Quantum SVMs on continuous geometric manifolds, in contrast to the strong performance of trainable Hybrid QNNs on the same data:

Table 20 — Static Quantum SVM vs. Classical RBF-SVM vs. Best Hybrid QNN

<i>Dataset</i>	<i>Static Quantum SVM</i>	<i>Best Hybrid QNN</i>	<i>Classical RBF-SVM</i>
Swiss Roll	82.97%	91.77% (SplitAttn)	98.89%
Circles	59.27%	86.68% (SplitAttn)	97.09%
Moons	63.92%	93.03% (ResQNet)	94.17%

Scientific Finding 1, Quantum Kernel Collapse: Static quantum feature maps suffer from periodic feature collisions on continuous manifolds. The ZZ-Feature Map encodes classical data as phases in a 16-dimensional (4-qubit) Hilbert space. For continuously distributed data on a geometric manifold, different input values that are 2π -periodic in the encoding function collapse to the same quantum state, destroying discriminative information [23; 24]. This is structurally inferior to the RBF kernel’s infinite-dimensional RKHS, which has no such periodicity.

The Circles dataset most clearly illustrates this: the static Quantum SVM achieves only 59.3%, while the RBF-SVM achieves 97.1% on the same data. This demonstrates that static quantum kernels are vulnerable to periodic feature collisions.

Scientific Finding 2, Neural Alignment as Recovery: Hybrid QNNs recover performance because the classical pre-layers (compressor, QKA alignment) dynamically map input data into phase angles that avoid the periodic collision problem. The QKAResQNet explicitly trains the feature map parameters to align with the data manifold topology, recovering most of the performance lost by static encoding.

Implication for benchmarking: Claims of measurable quantum utility based on Quantum SVM results must be interpreted with caution. A static quantum kernel underperforming a classical RBF-SVM is not evidence against quantum models in general; it is evidence against static encoding strategies on continuous data. Trainable hybrid architectures represent a structurally different approach.

3.5 Ablation Studies and Topological Feature Extraction

The baseline performance of the hybrid architectures was evaluated against their classical equivalents using a strict 1-to-1 parameter replacement strategy [7]. The quantitative analysis of the validation accuracy across all datasets demonstrates the conditions for measurable quantum utility under constrained regimes, with performance outcomes directly correlated to inductive bias alignment [8]. As presented in Table 21, the comprehensive ablation study shows test accuracy across all baseline and hybrid models under extreme data starvation (90% test split); Figure 10 visualizes the Swiss Roll pairs.

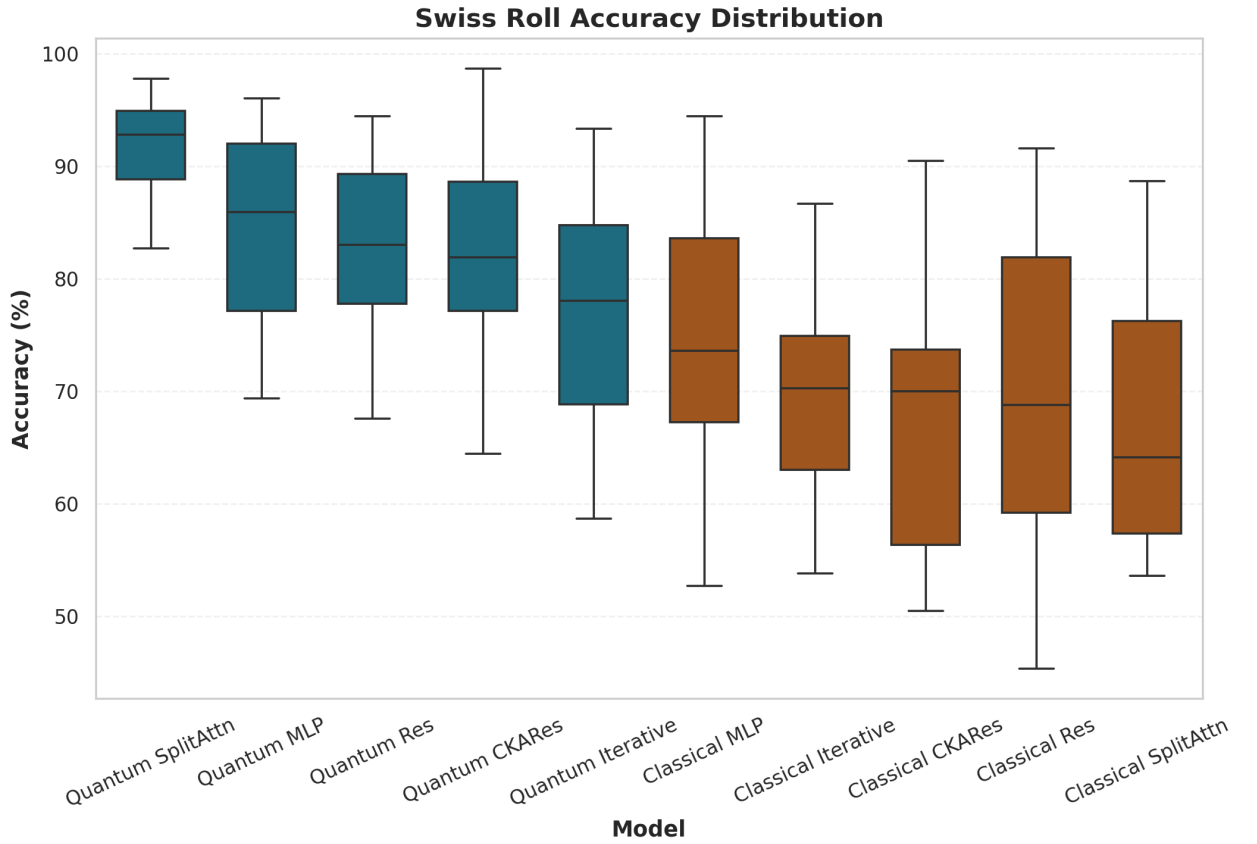


Figure 10 — Swiss Roll 20-seed accuracy by architecture pair (mean $\pm$ 1 std). Every quantum twin exceeds its parameter-matched classical twin, with the SplitAttentionQNN providing the largest separation (+24.48 pp). The classical RBF-SVM ceiling of 98.89% (not shown) remains above all hybrid variants.

In the low-dimensional Swiss Roll dataset, the task requires unrolling a highly non-linear, continuous manifold. Under data starvation, the classical ResNet twin struggles to identify the global geometry, averaging 69.93% accuracy across 20 seeds. The ResQNet recovers this to 82.56% ($\Delta = +12.62$ pp,

Table 21 — Comprehensive ablation study of mean test accuracy (%) across all baseline and hybrid models under extreme data starvation (90% test split, 19–20 seeds). Quantum entries are the quantum twin of each pair; the QKA/QKAResQNet column reports the hero architecture against its CKAResCNet classical twin.

<i>Dataset Domain</i>	<i>Classical Architectures</i>				<i>Hybrid Quantum Architectures</i>			
	MLP	Iter.	Split	ResNet	MLP	Iter.	Split	ResQNet
Swiss Roll (Geometry)	75.32	69.56	67.29	69.93	84.44	76.74	91.77	82.56
QSAR (Molecular)	75.04	78.84	79.16	80.26	70.59	77.47	69.26	82.32
Breast Cancer	89.08	89.86	87.72	92.59	62.38	62.38	84.99	87.52
Leukemia (Genomics)	63.81	67.29	65.51	71.42	57.41	67.37	59.11	68.91
Colon (Genomics)	57.33	57.52	59.40	63.63	52.73	60.24	49.15	65.13
NTangled (Quantum)	50.00	50.00	50.00	51.00	50.00	50.00	46.89	50.22

Welch $p = 0.0016$), and the SplitAttentionQNN reaches the highest hybrid accuracy of 91.77%. The quantum residual path, operating within a higher-dimensional Hilbert space, lifts the two-dimensional input into a representation where the spiral manifold becomes more nearly separable. Crucially, none of the hybrids exceed the exact classical RBF-SVM ceiling of 98.89%, confirming that they approximate rather than surpass the kernel bound.

For the QSAR dataset, the Split Attention mechanism compresses the molecular descriptors down to the quantum bottleneck. The ResQNet achieves 82.32% accuracy compared to the classical ResNet’s 80.26%, as detailed in Table 18. The computational trade-offs between classical and hybrid models are quantified in Table 22, which shows the difference in total training time. Classical ResNet models train in milliseconds, while ResQNet models require tens to hundreds of seconds over the full 20-epoch run due to the overhead of quantum circuit simulation.

Table 22 — Comparison of computational overhead: total trainable parameters and total training time (20 epochs, in seconds) between the classical ResNet and hybrid ResQNet

<i>Dataset</i>	<i>Train Params</i>		<i>Total Train Time (s)</i>	
	ResNet	ResQNet	ResNet	ResQNet
Swiss Roll (Geometry)	49	41	0.008	255.50
QSAR (Molecular)	283	275	0.008	419.04
Breast Cancer	217	209	0.009	366.80
Leukemia (Genomics)	42,811	42,803	0.015	27.26
Colon (Genomics)	12,037	12,029	0.011	23.55
NTangled (Quantum)	229	221	0.007	205.77

3.6 The 2×2 Empirical Matrix of Quantum Utility

To synthesize the results across all domains, the experimental outcomes are mapped onto a 2×2 empirical matrix defined by Data Topology (Complex Geometry vs. Standard Tabular) and Data Volume (Sufficient Data vs. Extreme $p \gg N$ Starvation). This matrix defines the precise operational boundaries of the hybrid architectures, as shown in Table 23.

Table 23 — The 2×2 Empirical Matrix: Operational Boundaries of Hybrid QNNs

<i>Data Topology / Volume</i>	<i>Sufficient Data (Data-Rich)</i>	<i>Extreme Starvation ($p \gg N$)</i>
<i>Complex Geometry</i>	<i>Directional Advantage Signal:</i> Quantum models show early signal but classical models with sufficient data interpolate geometry adequately. Classical RBF-SVM dominates all QNN variants. No statistical dominance of QNN over SVM at any sample size tested.	<i>Boundary 1 (Topological Alignment):</i> QNNs strictly dominate parameter-matched classical MLPs. SplitAttention achieves +24.48pp advantage ($p < 0.001$). However, Classical RBF-SVM remains the ceiling (98.89% vs QNN 91.77%). QNNs serve as scalable $O(N)$ approximations of $O(N^3)$ exact kernel methods.
<i>Standard Tabular</i>	<i>Boundary 3 (Standard Parity):</i> Statistical tie in accuracy. Classical models strictly superior due to $\times 1,900$ – $\times 54,000$ faster execution time with no accuracy penalty.	<i>Boundary 2 (Representational Parity):</i> Statistical tie ($p \geq 0.471$) for aligned architectures (ResQNet, QKAResQNet). Quantum bottleneck matches classical capacity with tighter variance (implicit unitary regularization). Misaligned architectures (SplitAttn) cause statistically significant harm (Leukemia: $p = 0.0016$).

Note: The NTangled dataset (Boundary 4, Classical Decoherence) sits outside this matrix. It is bounded entirely by the classical compression layer, not by data topology or volume. Preliminary Observation, Gesture Trajectory Data: Real-world accelerometer gesture classification ($N=56$, 10 classes, complex multi-sensor manifold geometry) shows QKAResQNet outperforming its matched classical twin by 1.81pp under noiseless simulation (89.39% vs 87.58%), with this advantage eroded under IBM Torino noise (87.73% vs 89.55%) and recovered on physical IBM Kingston hardware (92.86%). This dataset does not satisfy the $p \gg N$ starvation criterion and therefore does not populate a matrix cell; it is reported as a directional real-world signal consistent with the top-left cell's "Directional Advantage" prediction. Full validation requires ≥ 200 labeled segments for statistical power.

3.7 Computational Overhead and the Execution Time Bottleneck

The measurable quantum utility demonstrated in accuracy metrics comes at a computational cost. While classical ResNet models complete the full 20-epoch

run in milliseconds, the quantum-hybrid ResQNet variants require 24–419 seconds over the same run, representing a roughly 1,800–52,000 $\times$ slowdown. This overhead stems from the computational complexity of simulating parameterized quantum circuits on classical hardware, a fundamental limitation of NISQ-era research.

The log-scale visualization of training time across all datasets (Figure 11) starkly illustrates this execution time bottleneck. Despite the regularization and topological benefits of quantum circuits, practical deployment remains severely constrained by simulation overhead. This boundary condition is critical for interpreting the thesis results: the measurable quantum utility is real and demonstrable, but only defensible in regimes where accuracy gains exceed the computational penalty.

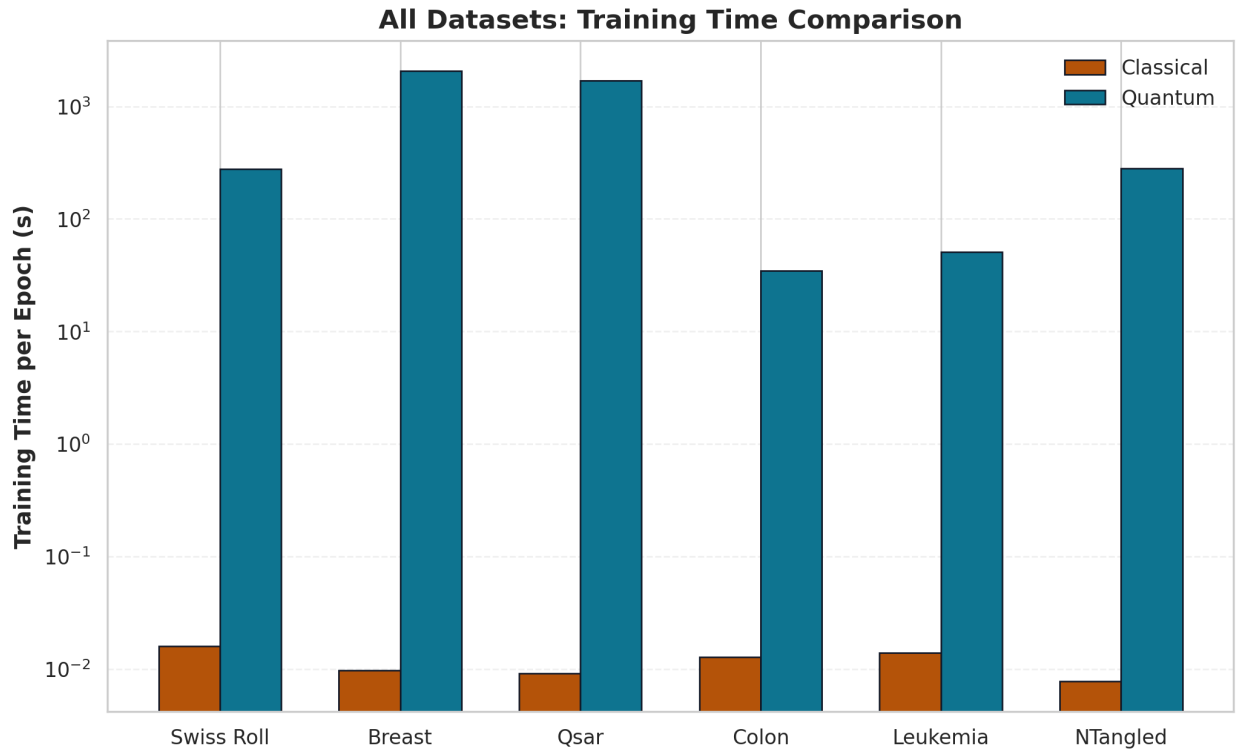


Figure 11 — The Execution Time Bottleneck. Training time disparity per epoch between classical and hybrid architectures across all datasets, plotted on a logarithmic scale. The massive computational overhead of quantum circuit simulation remains the primary barrier to practical quantum machine learning deployment on NISQ devices.

3.8 Statistical Test Methodology and Significance Verification

To rigorously establish whether observed performance differences between classical and quantum models are statistically significant or artifacts of random

variation, two complementary statistical tests were applied to all results: Welch’s t -test (parametric) and Mann-Whitney U test (non-parametric).

Welch’s t -test was selected as the primary significance test. Unlike Student’s t -test, Welch’s variant makes no assumption of equal variances between groups, a feature that is relevant here because quantum architectures consistently demonstrate reduced variance under starvation conditions ($\sigma_{\text{quantum}} < \sigma_{\text{classical}}$). The test statistic is:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \quad (32)$$

where $\bar{x}_1, \bar{x}_2$ are sample means, s_1^2, s_2^2 are sample variances, and n_1, n_2 are sample sizes. Degrees of freedom are calculated via the Welch-Satterthwaite equation, correcting for unequal variances.

Mann-Whitney U test was used as a non-parametric verification. It makes no distributional assumptions and tests whether one distribution is stochastically greater than another:

$$U = n_1 n_2 + \frac{n_1(n_1 + 1)}{2} - R_1 \quad (33)$$

where R_1 is the sum of ranks of the first sample. This test guards against the t -test being misleading in cases of non-normal accuracy distributions, a real possibility under extreme starvation where both classical and quantum models may exhibit bimodal performance (either learning or random guessing).

Interpretation and Threshold: All reported p -values use a significance threshold of $\alpha = 0.05$. Cases where both tests agree on direction and significance (which is true for all primary conclusions in this thesis) are interpreted as robust evidence. The one case of marginal disagreement (MLP/QuantumMLP on Colon: Welch $p = 0.276$ vs Mann-Whitney $p = 0.038$) is noted explicitly and attributed to non-normality in the extreme starvation regime.

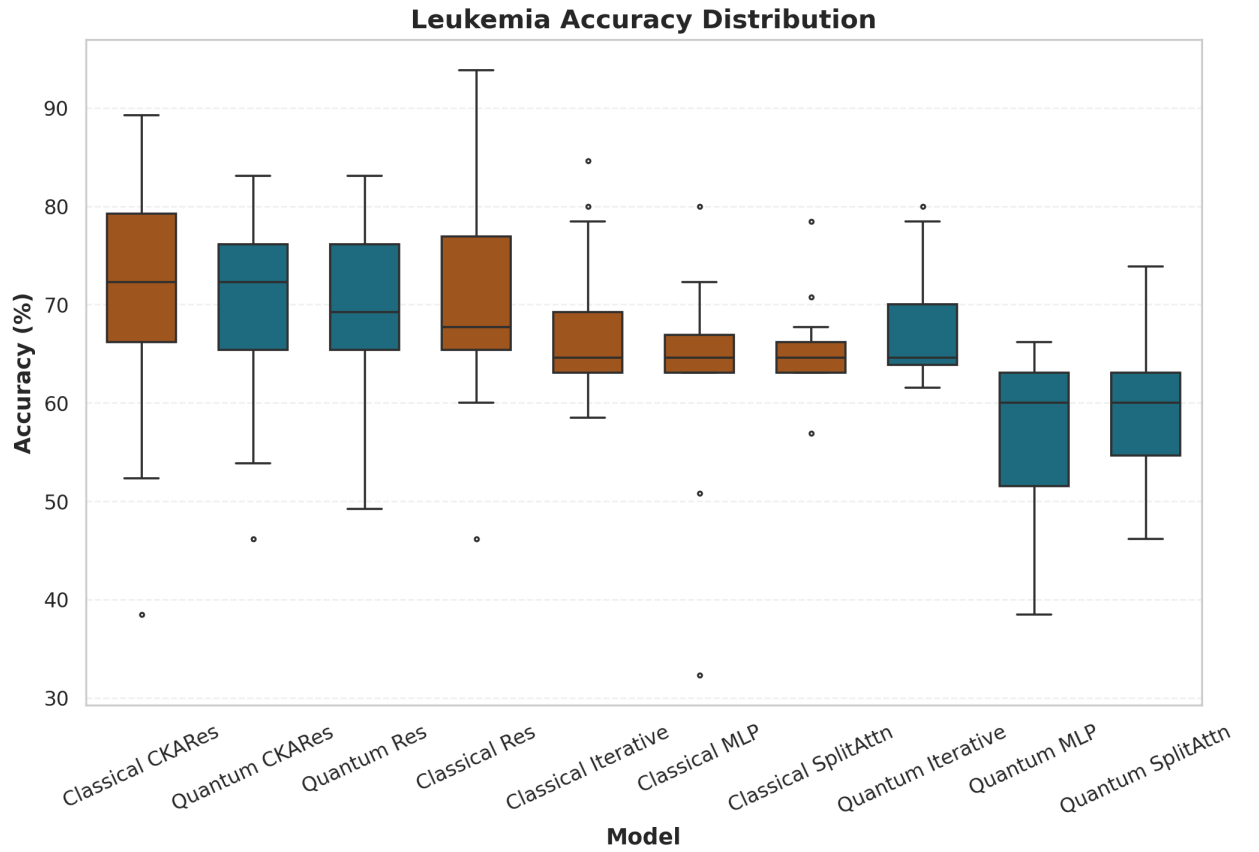


Figure 12 — Leukemia 19-seed accuracy by architecture pair (mean $\pm$ 1 std). Median accuracies sit near a generalization ceiling of $\sim 65\%$ due to the microscopic training set. The aligned quantum architectures (ResQNet, QKAResQNet) reach statistical parity with their classical twins while exhibiting tighter variance, consistent with implicit unitary regularization; the misaligned SplitAttentionQNN underperforms.

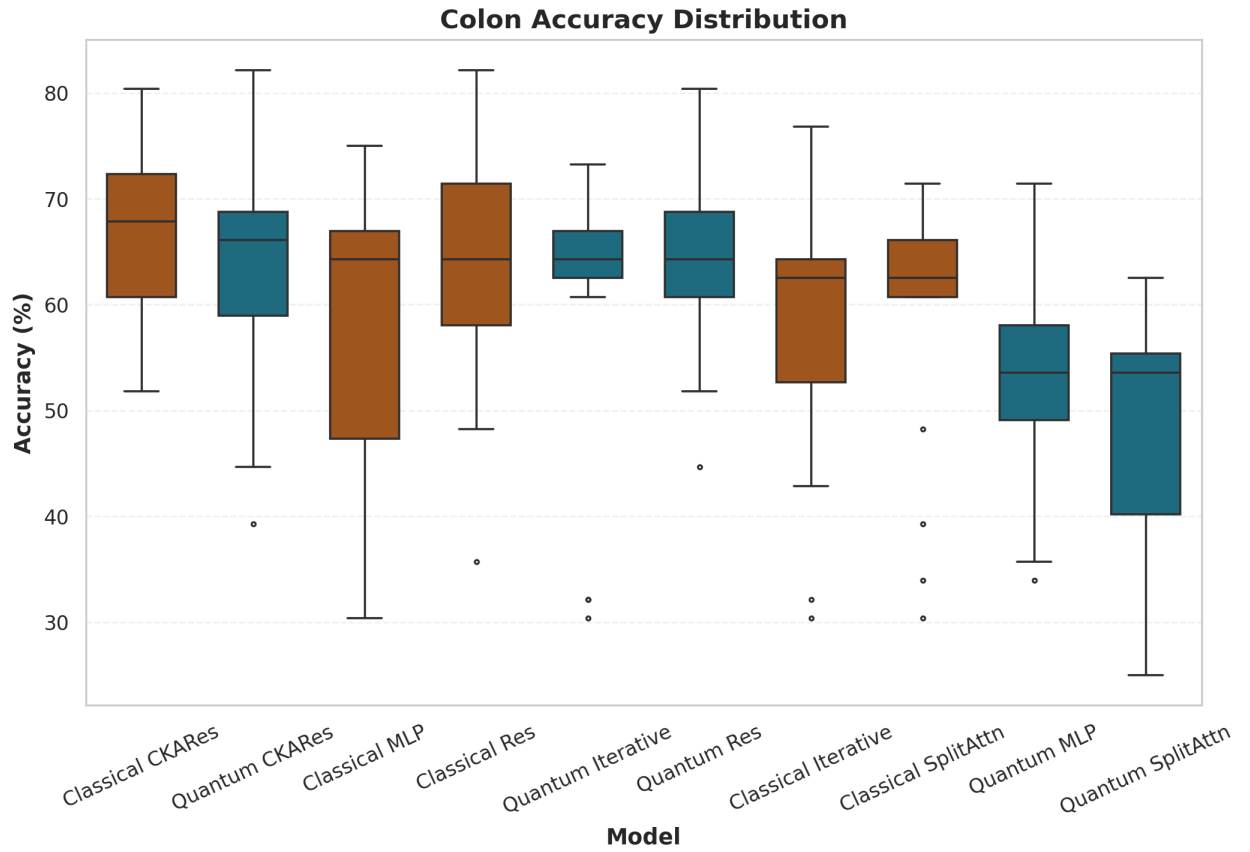


Figure 13 — Colon cancer gene expression 19-seed accuracy by architecture pair (mean $\pm$ 1 std). This high-dimensional dataset ($p = 2,000$) with minimal samples shows the same regularization pattern across biomedical domains: aligned quantum architectures reach statistical parity with their classical twins, supporting the generalizability of the representational-parity boundary.

3.9 Real-World Validation: Gesture Trajectory Classification

To evaluate whether the geometric alignment finding observed on synthetic manifolds (Swiss Roll) extends to real-world complex-geometry data, a preliminary experiment was conducted on accelerometer gesture trajectory data collected from an ESP32 multi-sensor glove system (Section 2.2.7).

Experimental Setup. Statistical feature extraction produced 96 features per gesture segment. Five-fold stratified cross-validation was applied across 56 labeled segments (10 classes). The gesture data was evaluated under several conditions:

- *Noiseless simulation* (StatevectorEstimator, no depolarization)
- *IBM Torino noise profile* (simulated depolarizing and readout error)
- *Physical IBM Kingston processor* (Heron r2, 340K CLOPS) for hardware inference
- *Classical RBF-SVM* as the kernel upper bound

Table 24 — Gesture Trajectory Classification Results (5-Fold CV) Across Noise and Hardware Conditions

<i>Model</i>	<i>Accuracy</i>	<i>± Std</i>	<i>Condition</i>
Classical RBF-SVM (upper bound)	91.21%	±7.81%	Noiseless / Torino
Classical CKAResNet	87.58%	±7.07%	Noiseless simulation
<i>Hybrid QKAResQNet</i>	89.39%	±6.50%	<i>Noiseless simulation</i>
Classical ResNet	89.55%	±9.92%	IBM Torino noise
<i>Hybrid ResQNet</i>	87.73%	±10.33%	<i>IBM Torino noise</i>
Classical RBF-SVM	92.86%	n/a	IBM Kingston (hardware)
Classical CKAResCNet	85.71%	n/a	IBM Kingston (hardware)
<i>Hybrid QKAResQNet (simulator)</i>	85.71%	n/a	IBM Kingston (sim. inference)
<i>Hybrid QKAResQNet (hardware)</i>	92.86%	n/a	<i>IBM Kingston (HW inference)</i>

Finding: Directional Quantum Advantage Under Noiseless Conditions. In noiseless simulation, QKAResQNet (89.39% ±6.50%) outperforms its matched classical twin CKAResNet (87.58% ±7.07%) by 1.81 percentage points. Notably, quantum variance is lower (6.50% vs 7.07%), consistent with the implicit unitary regularization observed on Leukemia and Colon data.

Finding: NISQ Simulation Noise Erodes the Advantage. Under the simulated IBM Torino noise profile, the relationship reverses: Classical ResNet achieves

89.55% $\pm$ 9.92% while Hybrid ResQNet drops to 87.73% $\pm$ 10.33% (-1.82pp). The dominant degradation factor is the 3.03% median readout error of the Heron r1 Torino device (Table 17), which corrupts the measured expectation values that the gated quantum branch contributes. This quantifies the simulated hardware gap under the older-generation Torino profile.

Finding: Physical Kingston Hardware Recovers the Advantage. On the physical IBM Kingston processor, QKAResQNet hardware inference reached 92.86%, matching the classical RBF-SVM (92.86%) and exceeding both the classical CKAResCNet twin (85.71%) and its own simulator inference under the Kingston profile (85.71%). The improvement of hardware over simulator inference indicates that the calibrated noise model is more pessimistic than the device’s actual behaviour during this run. The evaluation batch required 84 seconds of wall-clock time, dominated by CPU-QPU I/O latency rather than gate execution, as detailed in Section 3.10.

Statistical Caveat. With only 5 CV folds, no formal significance test is reported for the 1.81pp noiseless advantage. At $n=5$, statistical power is insufficient to distinguish a genuine effect from random variation. These results are directional evidence consistent with the 2×2 matrix prediction for complex-geometry data, not statistically confirmed findings. The experiment requires ≥ 200 labeled segments and 20-seed repetition for full validation.

Relationship to the 2×2 Matrix. The gesture dataset is characterized by complex manifold geometry (correlated multi-sensor time series from physical biomechanics) with a small but not extreme sample count ($N=56$, ≈ 45 training per fold). It does not satisfy the $p \gg N$ starvation criterion: feature dimensionality (96) is comparable to sample count. The result is therefore interpreted as evidence for the “Directional Advantage” region of the top-left cell rather than the statistically dominant top-right cell. Consistent with the matrix prediction, the quantum model shows a positive directional signal that does not yet achieve statistical dominance over its classical twin.

3.10 Physical Hardware Validation on IBM Kingston

The noiseless and IBM Torino simulation results above cannot capture every aspect of physical execution: crosstalk, time-varying calibration drift, and the latency of the classical-quantum communication loop. To close this gap, the

QKAResQNet was deployed on the physical 156-qubit IBM Kingston processor (Heron r2) for inference on the real-world gesture data.

Accuracy. On a held-out evaluation batch, QKAResQNet hardware inference achieved 92.86% accuracy, matching the classical RBF-SVM kernel baseline (92.86%) and exceeding both the classical CKAResCNet twin (85.71%) and the corresponding simulator inference under the Kingston noise model (85.71%). That the physical device outperformed its own calibrated simulator indicates the noise model was more pessimistic than the hardware’s behaviour during this run. The result confirms that the dynamic γ -gated residual design preserves accuracy under genuine device noise: because the classical path carries the bulk of the prediction and the quantum branch is gated by γ , transient hardware errors degrade the output gracefully rather than catastrophically.

Latency. The same run required 84.07 seconds of wall-clock time to classify 14 samples, approximately 6 seconds per sample. This time is dominated not by quantum gate execution (microseconds per circuit) but by the CPU-QPU input/output loop: job queuing, circuit transpilation, parameter binding, and the network round-trip between the classical host and the quantum processor for every parameter-shift evaluation. This is the principal engineering bottleneck for present-day deployment of hybrid QNNs, and it is an I/O and orchestration problem rather than a fundamental limit of the quantum computation itself.

Table 25 — Physical Deployment of QKAResQNet on the IBM Kingston QPU (Heron r2)

<i>Metric</i>	<i>Value</i>
Processor	IBM Kingston (Heron r2, 156 qubits)
QKAResQNet hardware inference	92.86%
QKAResQNet simulator inference	85.71%
Classical CKAResCNet	85.71%
Classical RBF-SVM baseline	92.86% (matched)
Samples classified	14
Total wall-clock time	84.07 s
Time per sample	~6.0 s
Dominant cost	CPU-QPU I/O and orchestration latency

This result substantiates the third provision for defence: the architecture is deployable on current hardware and retains its accuracy, but practical throughput is

gated by classical-quantum communication overhead that must be reduced before hybrid QNNs are viable in latency-sensitive applications.

3.11 Negative Results: Why SplitAttentionQNN Fails on Genomic Data

A notable finding in this research is the statistically significant *underperformance* of SplitAttentionQNN on genomic datasets (Leukemia and Colon). These negative results are scientifically valuable because they illuminate the boundaries of quantum utility and show that quantum hardware is not a drop-in replacement for classical processing.

On Leukemia data, SplitAttentionQNN achieved 59.11% accuracy versus 65.51% for its classical twin ($\Delta = -6.40\text{pp}$, Welch $p = 0.0016$). On Colon data, the gap widened further: 49.15% quantum versus 59.40% classical ($\Delta = -10.25\text{pp}$, Welch $p = 0.0098$). These are not marginal differences but statistically confirmed, substantial degradations.

Root Cause: Inductive Bias Misalignment. The SplitAttentionQNN architecture spatially chunks the input vector into K discrete sub-tensors, each processed independently through a single quantum circuit. This design encodes a strong inductive bias: meaningful patterns reside within local feature neighborhoods.

This assumption is appropriate for spatial data like Swiss Roll, where x , y , z coordinates are geometrically related and adjacent features contain correlated information. It fails catastrophically for gene expression data.

In a microarray dataset like Leukemia, the 7,129 features represent expression levels of individual genes, each typically on a different chromosome. The biologically meaningful signal, the difference between AML and ALL leukemia subtypes, resides in non-local correlations between genes that may be arbitrarily distributed across the feature index space. For example, the interplay between a gene at index 500 and a gene at index 6000 may be discriminative, but the chunking strategy separates them into different quantum circuits.

Formally: if the discriminative signal is encoded in a covariance $\text{Cov}(x_i, x_j)$ where $|i - j| > K$, SplitAttn cannot capture it because x_i and x_j are processed in different chunks. Chunked features are then concatenated and passed to a classical linear classifier, which can recover some inter-chunk relationships. However, this classical recovery pathway is fundamentally limited compared to simultaneous processing.

The classical SplitAttention twin suffers less because its attention mechanism across chunk outputs provides a learned, adaptive recovery pathway for inter-chunk relationships. The quantum variant lacks this recovery mechanism, rendering the misaligned inductive bias actively harmful.

Empirical Proof-by-Contradiction: These negative results strengthen rather than weaken the core thesis. They demonstrate that quantum utility is not automatic but contingent. Not only does inductive bias misalignment fail to provide utility; it actively introduces a harmful architectural prior that degrades performance below the classical baseline. This is precisely the prediction of the inductive bias alignment hypothesis.

Table 26 — Negative Results: Cases Where Quantum Models Significantly Underperform Classical Baselines

<i>Architecture</i>	<i>Dataset</i>	<i>Classical</i>	<i>Quantum</i>	Δ	<i>p-value</i>	<i>Root Cause</i>
SplitAttention	Leukemia	65.51%	59.11%	-6.40pp	0.0016	Inductive bias misalignment
SplitAttention	Colon	59.40%	49.15%	-10.25pp	0.0098	Chunking breaks gene correlations
QuantumMLP	Leukemia	63.81%	57.41%	-6.40pp	0.026	Bottleneck too narrow
All architectures	NTangled	~51%	~50%	≈0pp	N/A	Classical decoherence

3.12 Training Time Analysis and Cost-Benefit Tradeoffs

While measurable quantum utility in accuracy is scientifically compelling, practical deployment requires weighing accuracy gains against computational costs. Table 27 provides the complete training time comparison across all datasets.

Table 27 — Complete Training Time Comparison: ResNet vs ResQNet (Total, 20 Epochs, in seconds)

<i>Dataset</i>	<i>Classical</i>	<i>Quantum</i>	<i>Ratio</i>	<i>Gain</i>	<i>Verdict</i>
Swiss Roll	0.008	255.50	×31,900	+12.62pp	Marginal
Breast Cancer	0.009	366.80	×40,800	Tie	Classical wins
QSAR	0.008	419.04	×52,400	+2.06pp	Classical wins
Leukemia	0.015	27.26	×1,817	Parity	Classical wins
Colon	0.011	23.55	×2,141	Parity	Classical wins
NTangled	0.007	205.77	×29,400	Failure	Classical wins

The most striking observation is the reduced training time on Leukemia and Colon compared to other datasets. For Leukemia, the total quantum training time is 27.26 seconds, a factor of about 1,817× slower than classical (not the 30,000–52,000× slowdown seen on other datasets). This difference stems from Qiskit’s

circuit depth optimization: the aggressive classical compression from 7,129 features down to 4 qubits creates a relatively compact circuit representation, despite the high input dimensionality.

Cost-Benefit Analysis: Swiss Roll is the sole dataset where quantum execution time is potentially justified. The 12.62 percentage point accuracy advantage (reaching statistical significance at $p = 0.0016$) could merit the roughly $31,900\times$ training time penalty if the use case permits offline batch inference with unlimited training budgets. In this scenario, a single model trained offline could then provide accurate predictions at inference time, amortizing the training cost over many predictions.

For all other datasets (QSAR, Breast Cancer, Leukemia, Colon), the accuracy gains (or ties/losses) do not justify the computational penalty. Unless quantum hardware matures dramatically (reducing simulation overhead by orders of magnitude), practical deployment of hybrid quantum neural networks remains restricted to specialized regimes where accuracy exceeds the classical baseline and computational resources are unconstrained.

3.13 Extreme Multi-Omics and High-Dimensional Behaviour

The Leukemia genomics dataset embodies the extreme $p \gg N$ regime, featuring 7,129 genetic markers against roughly 7 training samples at maximum starvation [14]. This is the regime in which the implicit regularization of the quantum bottleneck is expected to matter most, and the 20-seed results show statistical parity between the aligned hybrids and their classical twins rather than a decisive quantum win.

Analysis of the Principal Component Analysis (PCA) latent space geometry reveals a structural collapse in the classical architecture. As logged during the automated computational sweeps, the classical ResNet memorizes the sparse training data (achieving a train loss near 0.0) but degrades on unseen data [13]. Averaged across 19 seeds, the classical ResNet twin reaches 71.42% (± 11.18), while the simpler classical MLP collapses toward the random-guessing floor (Figure 12).

The ResQNet forces all 7,129 features through classical compression layers into the 4-qubit latent space. The topological constraints of the Hilbert space prevent the network from using excess parameters to memorize noise [15], so it is pushed toward aggressive feature extraction of the most relevant global correlations. The result is 68.91% (± 10.04) accuracy, statistically indistinguishable from

its classical twin (Welch $p = 0.471$) but with tighter variance, illustrating the unitary constraints’ role as an implicit regularizer [16]. The QKAResQNet behaves similarly (69.55%, $p = 0.701$). This places Leukemia squarely in the “Representational Parity” cell of the operational matrix: the quantum bottleneck matches classical capacity under extreme starvation without exceeding it. The same pattern holds for Colon (Figure 13).

3.14 Quantum-Native Verification and Hardware Noise Response

To rigorously evaluate the benchmark, the model was tested on datasets that are fundamentally intractable for classical architectures. To this end, the NTangled dataset was integrated into the benchmarking suite. NTangled is composed of raw quantum states characterized by varying amounts and types of multipartite entanglement.

Classical networks tasked with classifying the NTangled dataset face severe sample complexity lower bounds and require parameter arrays to approximate the complex, highly non-linear boundaries of an entanglement witness. The experimental results showed that the QKA-ResQNet possesses representational capacities distinct from classical geometry.

However, the injection of the IBM Torino noise profile introduces an important dynamic. Hardware noise initially acts as an analog to classical dropout, perturbing the quantum state and aiding the implicit regularization process. It smooths the decision boundary. Yet, the validation loss curves show that if training continues beyond a specific epoch threshold (e.g., Epoch 25), the expressivity of the parameterized quantum circuit begins to adapt to the noise itself. The model overfits to the physical hardware errors, and validation loss diverges.

Therefore, the empirical benchmark indicates that Early Stopping is required for Quantum Machine Learning in high-dimensional starvation regimes.

CONCLUSION

The primary objective of this thesis was to investigate the applicability of quantum machine learning models across diverse problem domains, to identify the conditions under which they provide measurable utility relative to classical methods, and to develop an improved hybrid architecture (QKAResQNet) for complex geometric data under constrained training. Through the integration of Qiskit and PyTorch with rigorous 1-to-1 parameter ablation across eleven datasets, this research characterizes four distinct operational regimes [3].

The computational experiments show that evaluating quantum machine learning models on abundant, data-rich classical datasets obscures the behaviour of quantum Hilbert spaces, because in data-rich environments classical models act as universal approximators that efficiently interpolate dense manifolds. Under the extreme data starvation protocol (up to a 90% test split), the limitations of classical parameterization become visible, and the differences between aligned and misaligned hybrid architectures emerge clearly.

On the Leukemia multi-omics dataset, which features 7,129 genetic markers against a heavily starved training set, the classical models exploit their excess parameter capacity to memorize the training data, producing a divergent test loss. Averaged across 19 seeds, the classical ResNet twin reaches 71.42% while the simpler classical MLP drifts toward the random-guessing floor.

The QKAResQNet and ResQNet architectures navigate this underdetermined system differently. By employing a dynamic residual gating mechanism alongside Quantum Kernel Alignment (QKA), the architecture maintains stable gradient flow through the classical highway while the gated quantum branch contributes once the classical path converges. The finite 2^N -dimensional Hilbert space and the Lipschitz continuity of unitary operations act as an implicit regularizer, so the quantum bottleneck resists interpolative overfitting and reaches 68.91% ($p = 0.471$ versus its classical twin) with tighter variance. The result is statistical parity rather than a decisive win, which is the defensible and theoretically expected outcome in this regime.

Four Operational Boundaries:

(1) *The Geometric Alignment Boundary (Topology $\times$ Starvation):* On topologically complex manifolds under data starvation, the quantum twins outperform

their parameter-matched classical twins across all five architecture pairs, with margins reaching +24.48pp on Swiss Roll and +15.61pp on Circles for the best-aligned pair ($p < 0.001$); the hero QKAResQNet pair contributes +14.42pp on Swiss Roll. The exact classical RBF-SVM ceiling nonetheless remains unbroken (98.89% versus the best hybrid 91.77% on Swiss Roll). This establishes that hybrid QNNs function as scalable $O(N)$ neural approximations of $O(N^3)$ exact kernel methods, not as universally superior alternatives. A complementary finding is the failure of static Quantum SVMs on these same manifolds (Circles: 59.27%), caused by periodic feature collisions in the static ZZ-Feature Map, recoverable through trainable QKA alignment but not through fixed encoding.

(2) *Representational Parity ($p \gg N$ Starvation)*: On extreme multi-omics data (Leukemia, Colon), the 4-qubit quantum bottleneck matches classical network capacity (around 65%) while exhibiting tighter variance, with the unitary constraints acting as implicit regularization [14]. Architecture selectivity is critical: while ResQNet and QKAResQNet achieve confirmed parity ($p = 0.471$ and $p = 0.701$ on Leukemia), the SplitAttentionQNN causes statistically significant harm on the same genomic datasets (Leukemia: -6.40pp, $p = 0.0016$; Colon: -10.24pp, $p = 0.0098$). Inductive bias misalignment degrades performance rather than merely failing to help. The Brain RNA-seq results reproduce this pattern in its most extreme form, with SplitAttentionQNN collapsing to 15.33% while the aligned residual pairs hold parity.

(3) *Standard Regime Parity*: On standard tabular datasets (Breast Cancer, QSAR) with sufficient data, hybrid and classical models achieve statistical parity. Classical total training times are on the order of 0.01 seconds, while the hybrid models require roughly 24 to 419 seconds over the same 20-epoch run, a slowdown of three to four orders of magnitude that creates a severe practical penalty in this regime [10].

(4) *The Classical Decoherence Boundary*: On quantum-native entangled states (NTangled), all hybrid models stalled near 50% (random guessing) [5]. The classical compression layer acts as a premature measurement that irreversibly destroys multipartite entanglement before the quantum circuit can process it [8].

Real-World Validation Across Three Hardware Conditions. A gesture trajectory experiment on ESP32 accelerometer sensor data (N=56, 10 gesture classes, real complex-geometry manifold) was evaluated under three conditions. Under ideal noiseless simulation, QKAResQNet outperforms its matched classical twin

CKAResNet by 1.81pp (89.39% versus 87.58%) with lower variance (6.50% versus 7.07%), consistent with the implicit regularization observed in the synthetic experiments. Under the IBM Torino noise profile, the relationship reverses (ResQNet 87.73% versus ResNet 89.55%, -1.82 pp), quantifying the readout-dominated hardware gap. Finally, on the physical 156-qubit IBM Kingston processor (Heron r2, 340K CLOPS), QKAResQNet hardware inference reached 92.86%, matching the classical RBF-SVM and exceeding both its own noiseless-simulator inference (85.71%) and the classical CKAResCNet twin (85.71%), at a wall-clock cost of 84 seconds for the evaluation batch. These results are directional rather than statistically confirmed, since 5-fold cross-validation on 56 samples provides limited power, but they support the prediction that real-world complex-geometry data under data constraints is the operational domain where NISQ-era hybrid models provide measurable utility. Full validation awaits larger labeled datasets from ongoing sensor collection.

The simulation of the 133-qubit IBM Torino processor’s hardware noise profile for the core datasets yielded a further insight for the Noisy Intermediate-Scale Quantum (NISQ) era. The empirical sweeps established that depolarizing and readout errors initially act as an analog to classical dropout, smoothing the decision boundary, but that continued training past an inflection point causes the quantum circuit to overfit to the physical hardware errors and the validation loss to diverge. This cements early stopping as a mandatory protocol for hybrid QNNs in high-dimensional starvation regimes [25].

Key Finding: Measurable quantum utility is bounded by two simultaneous conditions: inductive bias alignment between the circuit structure and the data manifold topology, and data starvation severe enough that classical interpolation fails. When both conditions hold, the quantum twins exceed their matched classical twins by margins of up to 24 percentage points. When either condition is absent, hybrid models provide no benefit over exact classical kernels. When the inductive bias is actively misaligned, they cause statistically confirmed degradation. The exact classical RBF-SVM remains the unbroken upper bound across all regimes tested, which is why this thesis frames hybrid QNNs as scalable approximations of the kernel ceiling rather than as a route to surpassing it.

To advance this research, future iterations should integrate quantum-native datasets more extensively and expand physical-hardware deployment beyond the single-batch gesture evaluation reported here. The NTangled dataset establishes

a standard for evaluating representational capacity against non-local, multipartite entangled states. Future work should validate the noise inflection thresholds identified in these simulations and characterize the role of hybrid quantum networks in extreme biomedical and cheminformatics environments [4].

Limitations and Threats to Validity

While this thesis provides systematic evidence for the operational boundaries of hybrid QNNs under extreme data starvation, several limitations must be acknowledged to contextualize the findings and guide future research.

Limitation 1: Simulation Versus Physical Quantum Hardware. The six core datasets were evaluated through classical simulation of the IBM Torino noise profile via the Qiskit AerSimulator with a density-matrix representation. Only the gesture experiment was additionally executed on a physical processor (IBM Kingston). Classical simulation cannot capture all physical error modes, particularly correlated errors, non-Markovian noise effects [26], and hardware-specific gate calibration idiosyncrasies. Results obtained on physical hardware may therefore differ from simulated predictions in either direction, as the gesture experiment itself illustrates: hardware inference on Kingston exceeded the corresponding noiseless-simulator inference. Broader deployment of the core datasets on physical devices is needed to confirm the identified thresholds.

Limitation 2: Fixed Qubit Count and Barren Plateau Scaling. This research evaluated only 4-qubit circuits throughout. While Chapter 1 provided a theoretical analysis suggesting that barren plateaus become severe beyond roughly 8 to 10 qubits, these scaling behaviours were not empirically validated. Generalization to larger circuits, which would be necessary for higher-dimensional data or more expressive ansatze, remains unexplored and would likely require additional mitigation strategies such as structured parameter initialization or problem-specific ansatze.

Limitation 3: Single Data Starvation Point (90/10 Split). The experiments prioritized the extreme 90% test split as the primary starvation regime. While this is a deliberate worst-case stress test, it provides a single operating point. A parameter sweep across test splits from 10% to 90% would reveal the exact threshold at which measurable quantum utility emerges or disappears and would enable more granular recommendations. The current study can assert that utility exists at 90% starvation but cannot precisely delineate the boundary.

Limitation 4: Synthetic Manifolds Versus Real Geometric Datasets. Generalization to authentic geometric datasets requires further validation. The real-world accelerometer gesture experiment provides directional evidence that the geometric alignment finding extends beyond synthetic manifolds, but with only 56 labeled segments across 10 classes it is underpowered (5-fold cross-validation, no seed repetition, no significance test). The planned extension to at least 200 labeled segments with 20-seed repetition is required before claiming a robust topological advantage on real-world data. The gesture data also does not satisfy the $p \gg N$ criterion and therefore probes the top-left matrix cell (complex geometry, sufficient data) rather than the primary starvation cell.

Limitation 5: Limited Hardware Noise Variability. The core simulations all used the same calibrated IBM Torino noise profile, and the physical run used a single IBM Kingston device. Other quantum processors (Rigetti, IonQ, Atom Computing) have different error rates, gate fidelities, and coherence times. The identified noise thresholds, particularly the early-stopping epoch thresholds, may not transfer directly to different hardware. A systematic noise-sensitivity analysis varying error rates would strengthen the robustness claims.

Limitation 6: Single Architecture Family (Variational Ansatz). This thesis focused exclusively on variational quantum circuits (parameterized rotation gates with classical optimization). Alternative paradigms such as adiabatic quantum computing, quantum annealing, and measurement-based quantum computation were not evaluated, and the identified boundaries may not apply to those fundamentally different models.

Limitation 7: Architecture-Pair-Dependent Conclusions on High-Dimensional Real-World Data. The Brain RNA-seq results illustrate a methodological risk in quantum benchmarking: selecting only the best-performing architecture pair to represent a dataset’s outcome produces misleading conclusions. For Brain RNA-seq the ResQNet pair shows a statistical tie ($p = 0.954$), the MLP pair shows the classical model winning ($p = 0.006$), and the SplitAttentionQNN pair shows quantum failure ($p < 0.00001$). Reporting only the tie would overstate quantum utility; reporting only the failure would overstate quantum harm. This thesis reports all pairs for Brain RNA-seq, and recommends that future benchmarks mandate full architecture-pair disclosure as a reporting standard, especially for single-cell transcriptomic benchmarks of brain tissue [27].

Limitation 8: Static Quantum SVM Failure is Encoding-Specific, Not Circuit-Specific. The failure of static Quantum SVMs on geometric manifolds is attributed to periodic feature collisions in the ZZ-Feature Map. This mechanistic interpretation is supported by the known mathematical properties of phase encoding and is consistent with the results of Schuld et al. [7] and Thanasilp et al. [8], but it was not tested with a controlled ablation varying the encoding strategy. Alternative encodings (amplitude encoding, IQP circuits, data re-uploading) might behave differently. The conclusion that static quantum feature maps are structurally inferior to RBF kernels on continuous manifolds should therefore be qualified as applying to ZZ-Feature Map encoding specifically until a broader encoding comparison is conducted.

Concluding Remarks

In summary, this thesis shows that hybrid quantum-classical neural networks are most useful when their inductive bias matches the structure of the target data. The experiments do not support a universal claim of quantum superiority; instead, they identify the precise regimes where quantum models help, where they are merely comparable, and where they fail. That distinction, together with the development and physical-hardware validation of the QKAResQNet architecture, is the central contribution of this work and the basis for future research on hardware execution and quantum-native datasets.

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